SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

Design and Control of an Aerodynamic Surface-Enhanced Multirotor

William Constantin Rosenhahn

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I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, 01.12.2025 William Constantin Rosenhahn

Acknowledgments

"I would like to thank Pablo Kirby for developing the improved platform variant and supporting the experimental validation, and for the collaborative work environment."

Abstract

This thesis presents the design, modeling, and experimental validation of an X-wing hybrid aerial platform that combines quadrotor vertical take-off and landing (VTOL) capability with fixed-wing aerodynamic efficiency. The platform features four wings arranged in an X-configuration, providing passive lift that reduces required rotor thrust during forward flight.

A comprehensive dynamics model incorporating aerodynamic forces and moments is developed and integrated with an Incremental Nonlinear Dynamic Inversion (INDI) control architecture. The controller requires no explicit aerodynamic model and treats wing-generated forces as bounded disturbances, enabling robust trajectory tracking across the entire flight envelope from hover to high-speed cruise.

Experimental validation was conducted in a motion capture arena with a 2.5 kg prototype. Two configurations were tested: NACA 0015 airfoils at 45° dihedral and AG25 airfoils at 20° dihedral. Thrust characterization yielded the mapping $T = 5.15 \times 10^{-6} \cdot \mathrm{RPM}^2 - 0.090 \times 10^{-2} \cdot \mathrm{RPM}$ (N) with $R^2 = 0.997$. Flight tests demonstrated stable, accurate tracking up to $8\,\mathrm{m\,s^{-1}}$ with RMS position errors below 0.24 m, sustaining centripetal accelerations up to 2.8g, demonstrating stable control under strong unmodeled aerodynamic loads and indicative rotor unloading.

Electrical power was directly measured in outdoor flight using synchronized current and voltage telemetry. Results show a measurable reduction in mechanical power demand with increasing airspeed—up to 33% at $10\mathrm{m\,s^{-1}}$—confirming that aerodynamic lift from the fixed surfaces translates to tangible energy savings while maintaining full control authority.

Contents

1. Introduction

Conventional multirotors achieve unmatched agility and control simplicity but remain power-inefficient in sustained forward flight because all lift must be generated by rotor thrust. Fixed-wing aircraft, in contrast, exploit aerodynamic lift for high efficiency but cannot hover. Hybrid vertical-take-off-and-landing (VTOL) systems attempt to merge both regimes, yet they require additional actuators, complex transition control, and elaborate modeling.

This thesis investigates a minimal hybrid concept: a quadrotor augmented with fixed aerodynamic surfaces. The objective is to examine whether such a configuration can achieve measurable lift-induced power relief while retaining the full controllability of a multirotor and without increasing mechanical complexity.

1.1. Research question

The central research question is:

> *Can an Incremental Nonlinear Dynamic Inversion (INDI) controller maintain accurate trajectory tracking and flight stability on a quadrotor with significant aerodynamic lift, without relying on an explicit aerodynamic model, and can the resulting aerodynamic lift yield measurable electrical-power savings in forward flight?*

To answer this, the work combines modeling, design, and experimental validation. A six-degree-of-freedom dynamics model with simplified aerodynamic terms is derived. A fully 3D-printed "X-wing" quadrotor platform is built to match this model. The controller architecture employs a cascaded INDI structure with a geometric-based outer loop for position control and an inner INDI loop for attitude control, implemented in ROS 1 and validated through motion-capture experiments, as well as outdoor flights with synchronized power logging.

1.2. Contributions

The contributions of this thesis are:

- **Platform design**: A structurally symmetric X-wing quadrotor integrating four fixed wings into the rotor arms for passive lift generation.
- **Dynamics model**: A minimal 6-DoF model including quadratic lift and drag suitable for control and simulation.
- **Control architecture**: A cascaded INDI structure with a geometric-based outer loop for position control and an inner INDI loop for attitude control that compensates aerodynamic disturbances without explicit modeling.
- **Experimental validation**: Static thrust mapping with $R^2 = 0.997$ and indoor motion-capture trials and outdoor flights with synchronized power logging.
- **Efficiency assessment**: Measured rotor-power reduction up to 33% at $10\,\mathrm{m\,s^{-1}}$ compared with hover, confirming that passive lift directly reduces electrical power consumption.

These measurements provide the first quantitative demonstration of real-flight power savings on an aerodynamic surface-enhanced quadrotor under INDI control.

This investigation provides a reproducible reference for future research on aerodynamic surface-enhanced multirotors and hybrid UAV control without reliance on aerodynamic parameter identification.

1.3. Thesis organization

Chapter 2 reviews related work and motivates the chosen design. Chapter 3 derives the model. Chapter 4 details the platform. Chapter 5 presents the controller. Chapters 6 and 7 describe the setup and experiments. Chapter 8 concludes.

2. Related Work

The challenge of combining high agility with aerodynamic efficiency defines a long-standing trade-off in UAV design. Pure multirotors provide full six-degree-of-freedom control and extreme maneuverability but waste energy by tilting thrust for lift during forward flight. Fixed-wing UAVs achieve order-of-magnitude better cruise efficiency through aerodynamic lift but lose the ability to hover or hold arbitrary attitudes. Hybrid VTOL platforms attempt to bridge these regimes at the cost of additional actuators, transition logic, and increased control complexity.

2.1. Control versus Efficiency Trade-off

Multirotor control has matured through geometric and Incremental Nonlinear Dynamic Inversion (INDI) frameworks that achieve aggressive trajectory tracking without detailed modeling [SCM10; GLL15; Tzo+21]. These architectures rely solely on measured accelerations and remain robust under aerodynamic disturbances. However, they operate under the implicit assumption that thrust is the sole lift source—an assumption that breaks once fixed lifting surfaces contribute significant forces. The question is therefore whether these established control laws remain valid when lift is partly aerodynamic and unmodeled.

2.2. Existing Approaches and Their Limits

Hybrid VTOL concepts such as tilt-rotor and tailsitter aircraft achieve efficient cruise flight and transitions [TK22; Da24], yet their complexity is high. They require coordinated mode switching, non-linear actuator coupling, and precise aerodynamic modeling—elements that obscure the fundamental control question: can a standard multirotor controller tolerate significant aerodynamic forces without explicit modeling?

2.3. Aerodynamic Augmentation Without Added Complexity

Prior studies have explored fixed aerodynamic surfaces on multirotors as a simpler alternative. Dawkins and DeVries [DD18] demonstrated $\approx 35\,\%$ lower cruise power at 3–5 $\mathrm{m\,s^{-1}}$ with small attached wings, while Xiao et al. [XQa20] reported roughly 50 % electrical-power reduction at 15 $\mathrm{m\,s^{-1}}$ on a 1.2 kg platform, both without altering the baseline controller. These results suggest that passive aerodynamic surfaces can meaningfully reduce power consumption while maintaining full control authority. However, none of these works analyzed how unmodeled aerodynamic forces affect control stability or trajectory-tracking accuracy—especially under a disturbance-rejection scheme such as INDI.

2.4. Research Rationale

The literature therefore leaves an open gap: no prior study has systematically tested whether a disturbance-rejection controller such as INDI can maintain precision flight on a multirotor with strong, yet unmodeled, aerodynamic lift. Addressing this gap requires a configuration that introduces aerodynamic coupling without adding mechanical or algorithmic complexity. A quadrotor with fixed wings—an aerodynamic surface-enhanced multirotor—satisfies that criterion.

Consequently, this thesis focuses on (i) implementing an INDI-based controller on such a platform, (ii) quantifying its tracking accuracy under aerodynamic loading, and (iii) evaluating whether explicit aerodynamic modeling is unnecessary within the tested speed envelope.

2.5. Summary and Transition

Existing research establishes the energy-efficiency benefits of fixed aerodynamic surfaces on multicopters but leaves their influence on control robustness largely unexplored. The missing link is a quantitative examination of how a conventional multirotor controller—specifically, an INDI-based architecture—behaves when aerodynamic lift and drag become significant yet remain unmodeled.

To address this, the present work isolates the essential physics by selecting the simplest possible testbed: a standard quadrotor geometry augmented with symmetric fixed wings. The next chapter formalizes its six-degree-of-freedom dynamics and defines the aerodynamic forces that the controller will deliberately ignore. This provides the analytical baseline against which the later experimental results can be interpreted.

3. Dynamics Model

The following model provides a formal reference for all forces and moments acting on the vehicle but is not employed directly by the controller. The purpose is two-fold:

1. To define the dynamic relationships and notation used in subsequent analysis and simulation.
2. To quantify the order of magnitude of aerodynamic forces that the INDI controller must reject without an explicit model.

The controller described in Chapter 5 relies solely on incremental acceleration feedback; none of the aerodynamic coefficients or equations derived here are required in real time.

3.1. List of Symbols

Table 3.1.: Symbols used in the dynamics model.

Symbol	Description	Unit
$\mathcal{W}$	World (inertial) frame (ENU)	—
$\mathcal{B}$	Body-fixed frame at CoG	—
$\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$	Body-frame unit vectors	—
$\mathbf{p}$	Position in world frame	m
$\mathbf{v}$	Velocity in world frame	m/s
R	Rotation matrix $\mathcal{B} \to \mathcal{W}$	—
$\boldsymbol{\omega}$	Angular velocity in body frame	rad/s
$\mathbf{g}$	Gravity vector $[0, 0, -g]^\top$	m/s^2
m	Vehicle mass	kg
J	Inertia matrix in body frame	kg·m^2
$\mathbf{F}_g$	Gravitational force	N

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Table 3.1 – continued from previous page

Symbol	Description	Unit
f_i	Thrust produced by rotor i	N
$\mathbf{u}$	Control input vector $[f_1, f_2, f_3, f_4]^\top$	N
ω_i	Angular speed of rotor i	rad/s
k_t	Thrust coefficient	N·s^2·rad^{-2}
b_i	Static bias in thrust model for rotor i	N
k_Q	Torque coefficient	N·m·s^2·rad^{-2}
Q_i	Reaction torque of rotor i	N·m
c_i	Prop rotation sign (± 1)	—
$\mathbf{v}_w$	Wind velocity in world frame	m/s
$\mathbf{v}_a$	Airspeed in body frame	m/s
V	Airspeed magnitude	m/s
α	Angle of attack	rad
β	Sideslip angle	rad
ρ	Air density	kg/m^3
q	Dynamic pressure $= \frac{1}{2}\rho V^2$	N/m^2
S	Wing reference area	m^2
b	Wing span	m
AR	Aspect ratio b^2/S	—
a_α	Lift-curve slope	rad^{-1}
a_β	Side-force slope (or C_{Y_β})	rad^{-1}
C_L, C_D, C_Y	Lift, drag, side-force coefficients	—
C_{D0}	Zero-lift drag coefficient	—
k	Induced drag factor	—
$\mathbf{F}$	Total force in world frame	N
$\mathbf{F}_T$	Rotor thrust force in world frame	N
$\mathbf{F}_{aero}$	Aerodynamic force in body frame	N
$\boldsymbol{\tau}$	Total moment in body frame	N·m
$\boldsymbol{\tau}_T$	Rotor-generated moment	N·m
$\boldsymbol{\tau}_{aero}$	Aerodynamic moment	N·m
$\mathbf{r}_i$	Arm vector to rotor i	m
$\mathbf{r}_w$	Aerodynamic center offset	m
$\hat{\mathbf{t}}_i$	Rotor i thrust direction unit vector	—
G	Allocation matrix (force–moment mapping)	—
$\hat{\mathbf{v}}$	Airspeed unit vector	—

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Table 3.1 – continued from previous page

Symbol	Description	Unit
$\hat{\mathbf{z}}_L$	Lift direction unit vector	—
$\hat{\mathbf{y}}_S$	Side-force direction unit vector	—

3.2. Frames, States, Inputs

We use a world (inertial) frame $\mathcal{W}$ with ENU convention and a body-fixed frame $\mathcal{B}$ at the CoG. Gravity is $\mathbf{g} = [0,\, 0,\, -g]^\top$ in $\mathcal{W}$.

State:

$$(\mathbf{p}, \mathbf{v}, R, \boldsymbol{\omega}) \in \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3) \times \mathbb{R}^3,$$

where $\mathbf{p}, \mathbf{v}$ are position/velocity in $\mathcal{W}$, R maps $\mathcal{B} \to \mathcal{W}$, and $\boldsymbol{\omega}$ is the body angular rate in $\mathcal{B}$.

Control input (per rotor $i \in \{1, \dots, 4\}$):

$$\mathbf{u} = [f_1,\, f_2,\, f_3,\, f_4]^\top, \qquad f_i \geq 0,$$

with nominal thrust along $+\mathbf{e}_3$ of $\mathcal{B}$. Motor dynamics and allocation are covered in Chapter 5.

Wind and airspeed. Let $\mathbf{v}_w \in \mathbb{R}^3$ be the wind in $\mathcal{W}$. Body-frame airspeed:

$$\mathbf{v}_a \;=\; R^\top(\mathbf{v} - \mathbf{v}_w) \;\in\; \mathbb{R}^3, \quad V \;=\; \|\mathbf{v}_a\|.$$

Define angle of attack and sideslip (aircraft convention, small angles):

$$\alpha \;=\; \operatorname{atan2}\bigl(-v_{a,z},\, v_{a,x}\bigr), \qquad \beta \;=\; \arcsin\Bigl(\frac{v_{a,y}}{V}\Bigr),$$

so that $\alpha \approx -v_{a,z}/v_{a,x}$ and $\beta \approx v_{a,y}/V$ for $|v_{a,z}| \ll |v_{a,x}|$, $|v_{a,y}| \ll V$.

3.3. Forces in the World Frame

Total force:

$$\mathbf{F} \;=\; m\mathbf{g} \;+\; \mathbf{F}_T \;+\; R\,\mathbf{F}_{aero}. \tag{3.1}$$

3.3.1. Gravity

$\mathbf{F}_g = m\mathbf{g}$.

3.3.2. Rotor Thrust

Each rotor i produces a thrust f_i along its (possibly tilted) direction $\hat{\mathbf{t}}_i$ in $\mathcal{B}$. The net thrust in $\mathcal{W}$ is

$$\mathbf{F}_T = R\sum_{i=1}^{4} f_i\,\hat{\mathbf{t}}_i. \tag{3.2}$$

For small fixed tilts (e.g., $\pm 5°$) we also provide the common approximation $\mathbf{F}_T \approx (\sum_i f_i)\,R\mathbf{e}_3$; all tilt effects are embedded in the allocation matrix G used for moments.

3.3.3. Aerodynamic Forces on Fixed Surfaces

We use a lumped, quasi-steady wing model. Dynamic pressure $q = \frac{1}{2}\rho V^2$, total reference area S.

Lift/drag coefficients:

$$C_L(\alpha) = a_\alpha\,\alpha, \qquad C_D(\alpha) = C_{D0} + k\,C_L(\alpha)^2.$$

The lift coefficient C_L is modeled as a linear function of the angle of attack α:

$$C_L(\alpha) = a_\alpha\,\alpha, \tag{3.3}$$

where a_α [rad^{-1}] is the lift-curve slope. C_L represents the dimensionless ratio of lift force to dynamic pressure and reference area, $C_L = \frac{L}{\frac{1}{2}\rho V^2 S}$. The angle of attack α is defined as the angle between the incoming airflow and the vehicle body x-axis (or the mean aerodynamic chord of the wing). For thin airfoils and small angles, a_α can be approximated by 2π rad^{-1} in 2D, or by the finite-wing correction $a_\alpha = \frac{2\pi AR}{AR+2}$ with aspect ratio $AR = b^2/S$. We measure α relative to the body x-axis (mean aerodynamic chord aligned with $\mathbf{e}_x$); the linear model holds for attached flow ($|\alpha| \lesssim 10°$). This linear relation is valid up to moderate angles and forms the basis for the simplified quadratic aerodynamic model used here.

Side-force convention. We use $C_Y(\beta) = C_{Y_\beta}\,\beta$ with C_{Y_β} [rad^{-1}] the stability derivative. For conventional weathercock-stable configurations $C_{Y_\beta} < 0$ (side-force opposes β). In our small UAV geometry C_{Y_β} is identified experimentally and may be small in magnitude.

Optional lateral (side-force) coefficient:

$$C_Y(\beta) = a_\beta\,\beta.$$

Aerodynamic axes. We assume the wing span aligns with $\mathbf{e}_y$, so lift lies in the (x, z) plane. The definitions of $\hat{\mathbf{z}}_L$ and $\hat{\mathbf{y}}_S$ are undefined when $\hat{\mathbf{v}}$ is parallel to the respective cross-product axis (e.g., $\hat{\mathbf{v}} \parallel \mathbf{e}_y$ or $\hat{\mathbf{v}} \parallel \mathbf{e}_z$); in implementation we fall back to the last valid direction or a small-angle approximation.

Unit aerodynamic directions in $\mathcal{B}$ (from $\mathbf{v}_a$):

$$\hat{\mathbf{v}} = \frac{\mathbf{v}_a}{V}, \quad \hat{\mathbf{z}}_L = \frac{\hat{\mathbf{v}} \times \mathbf{e}_y}{\|\hat{\mathbf{v}} \times \mathbf{e}_y\|}, \quad \hat{\mathbf{y}}_S = \frac{\mathbf{e}_z \times \hat{\mathbf{v}}}{\|\mathbf{e}_z \times \hat{\mathbf{v}}\|},$$

so that lift is perpendicular to the flow in the body x–z plane, drag opposes $\hat{\mathbf{v}}$, and side force is lateral.

Aerodynamic force in $\mathcal{B}$:

$$\mathbf{F}_{aero} = \underbrace{qS\,C_L(\alpha)\,\hat{\mathbf{z}}_L}_{\text{lift}} - \underbrace{qS\,C_D(\alpha)\,\hat{\mathbf{v}}}_{\text{drag}} + \underbrace{qS\,C_Y(\beta)\,\hat{\mathbf{y}}_S}_{\text{side}}. \tag{3.4}$$

The aerodynamic forces in (3.4) are used only for off-line analysis; they are not used in the controller.

Controller relevance. These aerodynamic forces are *not* modeled in the controller; they are treated as disturbances. INDI cancels their quasi-steady contribution via incremental updates (Chapter 5).

3.4. Moments in the Body Frame

Total moment:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_T + \boldsymbol{\tau}_{aero}. \tag{3.5}$$

3.4.1. Rotor-Generated Moments

Let $\mathbf{r}_i$ be the arm vector from CoG to rotor i, and $c_i \in \{+1, -1\}$ the prop rotation sign (reaction torque direction). Here $c_i \in \{+1, -1\}$ encodes the rotor spin direction (positive for CCW viewed from above). With small motor tilt handled by per-rotor thrust directions $\hat{\mathbf{t}}_i$,

$$\boldsymbol{\tau}_{\text{arms}} = \sum_{i=1}^{4} \mathbf{r}_i \times (f_i\,\hat{\mathbf{t}}_i), \tag{3.6}$$

$$\boldsymbol{\tau}_{\text{react}} = \sum_{i=1}^{4} c_i\,Q_i\,\mathbf{e}_3, \qquad Q_i \approx k_Q\,\omega_i^2, \tag{3.7}$$

so that $\boldsymbol{\tau}_T = \boldsymbol{\tau}_{\text{arms}} + \boldsymbol{\tau}_{\text{react}}$. In nominal modeling $\hat{\mathbf{t}}_i = \mathbf{e}_3$; with $\pm 5°$ motor tilt we precompute $\hat{\mathbf{t}}_i$ and absorb it in the allocation matrix G (see Chapter 5).

3.4.2. Aerodynamic Moments

Wing forces act at an effective aerodynamic center offset $\mathbf{r}_w$ (in $\mathcal{B}$). We lump unsteady effects into a disturbance:

$$\boldsymbol{\tau}_{aero} \;=\; \mathbf{r}_w \times \mathbf{F}_{aero} \;+\; \boldsymbol{\tau}_{aero,dist}. \tag{3.8}$$

As with forces, the controller does not model $\boldsymbol{\tau}_{aero}$ explicitly; INDI compensates the quasi-steady part incrementally.

3.5. Equations of Motion

Rigid-body dynamics:

$$\begin{aligned} \dot{\mathbf{p}} &= \mathbf{v}, && (3.9)\\ \dot{\mathbf{v}} &= \tfrac{1}{m}\big(m\mathbf{g} + \mathbf{F}_T + R\,\mathbf{F}_{aero}\big), && (3.10)\\ \dot{R} &= R[\boldsymbol{\omega}]_\times, && (3.11)\\ J\dot{\boldsymbol{\omega}} &= -\boldsymbol{\omega} \times J\boldsymbol{\omega} + \boldsymbol{\tau}, && (3.12) \end{aligned}$$

with inertia J in $\mathcal{B}$ and $[\cdot]_\times$ the skew operator.

3.6. Thrust and Allocation (Static Mapping)

For simulation and identification we use static rotor maps

$$f_i \;=\; k_t\,\omega_i^2 + b_i, \qquad Q_i \;=\; k_Q\,\omega_i^2,$$

and a (geometry-dependent) allocation matrix $G \in \mathbb{R}^{4\times 4}$ such that

$$\begin{bmatrix} f_T \\ \boldsymbol{\tau} \end{bmatrix} \;=\; G\,\mathbf{f}, \qquad \mathbf{f} = [f_1, f_2, f_3, f_4]^\top, \quad f_T = \mathbf{1}^\top\mathbf{f}.$$

Motor tilt is embedded in G via $\hat{\mathbf{t}}_i$ and the arm geometry. Chapter 5 details the inverse mapping and constraints.

3.6.1. Rotor Power Scaling

Understanding how rotor power depends on rotational speed is essential for evaluating the efficiency of hybrid multirotor–fixed-wing configurations. In hover, this relationship can be derived from momentum theory, which models the rotor as an ideal actuator disk accelerating air downward to produce thrust.

For a rotor in hover, the thrust T is generated by imparting a downward momentum to the air mass flow $\dot{m}$:

$$T = \dot{m}v_i = (\rho A v_i)v_i = \rho A v_i^2,$$

where ρ is the air density, A is the rotor disk area, and v_i is the induced velocity through the disk. The corresponding mechanical power required to sustain this thrust is

$$P = Tv_i = \rho A v_i^3.$$

Eliminating v_i using the first equation yields the classical induced power expression from actuator-disk theory:

$$P = T\sqrt{\frac{T}{2\rho A}} = \frac{T^{3/2}}{\sqrt{2\rho A}}$$

(see [Fed23]).

If blade geometry and collective pitch remain constant, blade-element theory shows that the thrust coefficient C_T remains approximately constant, implying

$$T \propto \Omega^2,$$

where Ω is the rotor angular velocity. Substituting this into the induced power relation gives the cubic power law

$$P \propto \Omega^3.$$

Thus, reducing rotor speed by 20% results in a power reduction of roughly $(0.8)^3 \approx 0.51$, or about 49% lower power consumption—an important efficiency benefit when part of the lift is provided by wings.

This relationship strictly applies to hover or near-hover conditions, assuming uniform inflow and negligible profile and parasitic losses. In forward or edgewise flight, blade-element momentum theory (BEMT) or CFD analyses are required for accurate power prediction, but the cubic scaling remains a useful first-order engineering approximation for small deviations around hover.

The cubic law provides a theoretical baseline later compared against measured electrical-power data from outdoor experiments (Chapter 7), enabling direct verification of aerodynamic efficiency gains.

3.7. Assumptions and Simplifications

- Quasi-steady aerodynamics.
- No explicit prop-wash/wing interference; its net effect is absorbed in the disturbance terms and identified coefficients.

- Rigid airframe, symmetric mass properties about $\mathcal{B}$ axes.
- Fixed-pitch rotors; flapping and gyroscopic coupling are neglected in the baseline model.
- Coefficient models $C_L = a_\alpha \alpha$, $C_Y = a_\beta \beta$, $C_D = C_{D0} + kC_L^2$ valid for the moderate α, β range used here.

3.8. Summary and Outlook

The presented six-degree-of-freedom equations define a consistent reference for the vehicle's translational and rotational dynamics. Aerodynamic terms were included only to quantify expected disturbance magnitudes—order of lift, drag, and pitching moments—so that the later control analysis can assess whether these effects are sufficiently slow and quasi-steady to be rejected by INDI. No aerodynamic coefficients or force models are used online; they serve purely for interpretation and comparison.

With the governing relationships and notation established, the next step is to translate this abstract model into a physical platform whose geometry and mass distribution match the assumed parameters. The following chapter therefore details the mechanical design and manufacturing of the X-wing quadrotor, showing how the chosen dimensions, airfoil, and dihedral angles realize the modeled reference frame in practice.

4. Platform Design

The platform was designed to isolate the aerodynamic effects of fixed lifting surfaces on a quadrotor without introducing new actuators or changing the control structure. The result is an X-wing configuration in which each rotor arm forms a fixed lifting surface. The design goal was to achieve measurable lift at moderate forward speeds (5–15 $\mathrm{m\,s^{-1}}$) while preserving the agility and controllability of a conventional multirotor.

4.1. Configuration and Layout

The vehicle retains the standard X-shaped quadrotor geometry but extends each arm into a symmetric wing section, producing a cross-shaped planform. This geometry satisfies three constraints:

- **Dynamic symmetry**: Equal inertias and aerodynamic properties in roll and pitch enable identical control gains across both axes.
- **Passive lift generation**: Each wing contributes lift proportional to its forward velocity component without interfering with rotor control authority.
- **Mechanical simplicity**: No additional servos or morphing structures are introduced; all aerodynamic effects are passive.

The $\pm 45°$ alternating dihedral and anhedral angles maintain directional symmetry, ensuring that any heading can serve as "forward" flight. Although such extreme dihedral is aerodynamically suboptimal compared with planar wings, it simplifies the control problem by avoiding preferred directions and limiting coupling between roll and pitch.

The wings employ a NACA 0015 symmetric profile with 0.25 m chord and 0.45 m span, yielding an aspect ratio of ≈ 1.8. These dimensions were selected to ensure that at $\approx$ 10–12 $\mathrm{m\,s^{-1}}$ the predicted lift equals or slightly exceeds the vehicle weight, based on first-order thin-airfoil theory. The design therefore enables significant lift without requiring impractically large structures.

4.2. Structural Design and Manufacturing

Each wing consists of a 3D-printed aerodynamic shell reinforced by two 10 mm carbon spars. The hybrid construction provides sufficient bending stiffness while keeping individual wing mass below 180 g. Structural resonance tests indicated the first bending mode well above the control bandwidth, confirming negligible coupling to attitude dynamics.

The central frame clamps the eight spars from the four wings and houses all electronics. Motors are mounted on integrated endcaps that also serve as landing feet, providing 5° outward motor tilt for yaw authority. The total diagonal motor-to-motor distance is 1.1 m, yielding a compact yet aerodynamically relevant planform.

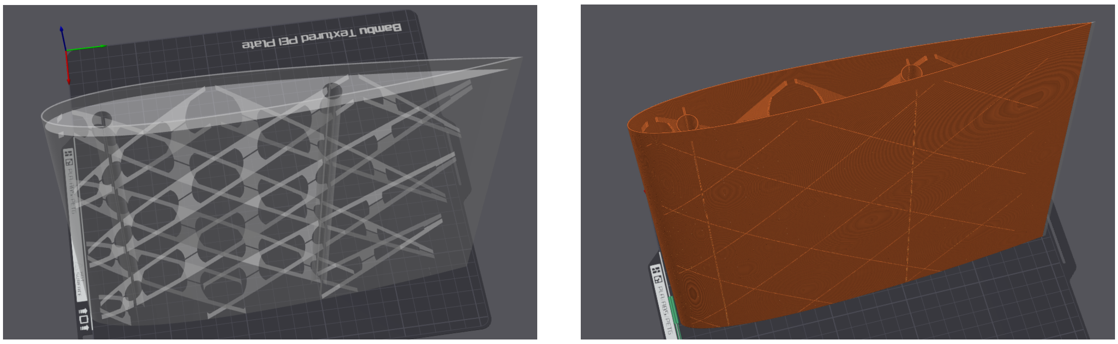

Figure 4.1.: Wing structure showing the hollow internal geometry with channels for carbon fiber spars (left) and printed aerodynamic shell (right). Each wing weighs approximately 176 g including carbon reinforcement.

4.3. Aerodynamic Estimates

At $12\,\mathrm{m\,s^{-1}}$, using $\rho = 1.225\,\mathrm{kg\,m^{-3}}$, $S = 0.318\,\mathrm{m^2}$, $C_L = 0.97$, $C_D = 0.025$, the theoretical lift and drag are:

$$L \approx 27.2\,\mathrm{N}, \quad D \approx 0.7\,\mathrm{N}.$$

This corresponds to $\approx 110\,\%$ of the vehicle's 2.5 kg weight, suggesting that in steady forward flight the rotors primarily counteract drag and provide control authority. Although these values are first-order estimates and neglect prop-wash interaction, they provide a reference for disturbance magnitudes that the INDI controller must compensate.

More accurate modeling would require CFD or wind-tunnel validation; however, for the scope of this study the analytical approximation suffices to show that aerodynamic

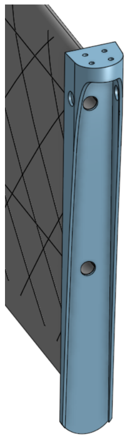

Figure 4.2.: Wing endcap component serving as motor mount and landing foot. This multi-functional part positions the motor ahead of the leading edge with $\pm 5^{\circ}$ tilt for yaw control and extends downward to protect the wing trailing edge during landing.

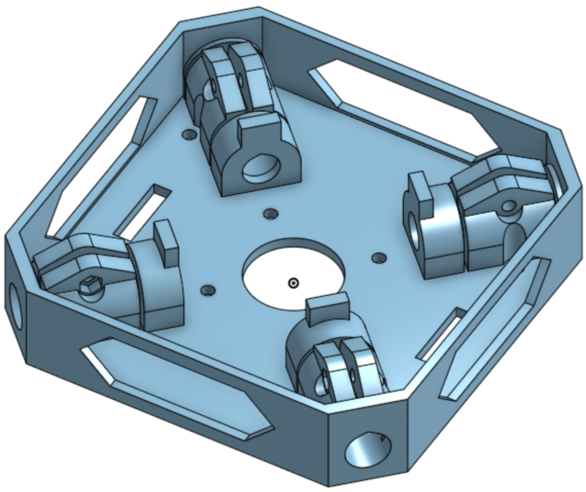

Figure 4.3.: Central core structure that clamps the carbon fiber spars. Two identical cores are used: one houses the ESC, flight controller, and batteries (top), while the other serves as a structural clamp for the lower spars (bottom).

lift is on the same order as vehicle weight and therefore cannot be ignored in flight analysis, even if it remains unmodeled in control.

A detailed comparison of the NACA 0015 baseline airfoil against an improved AG25 airfoil, including aerodynamic performance metrics and their correlation with flight test results, is presented in Chapter 7.

4.4. Propulsion and Actuation

The propulsion system uses T-MOTOR VELOX V2808 motors with HQProp 7×3.5×3 propellers, producing up to 22 N static thrust per motor. The total thrust-to-weight ratio exceeds 3, allowing aggressive maneuvers and ensuring that aerodynamic loads never saturate actuator authority.

Motors are tilted outward by 5° to enhance yaw torque. The corresponding geometry is embedded in the allocation matrix used by the controller; no additional coupling compensation is required. Static thrust characterization (Chapter 7, Section 7.1) confirms a quadratic thrust–RPM relation with $R^2 = 0.997$, validating consistency between analytical assumptions and measured performance.

The complete airframe mass, including avionics and batteries, is ≈ 2.5 kg. The center of gravity lies at the geometric center, aligned with the intersection of the four wing roots, minimizing trim moments during hover.

4.5. Summary

The X-wing configuration provides a mechanically simple and dynamically symmetric platform in which aerodynamic forces are large enough to challenge the controller but not strong enough to destabilize the system. The design therefore fulfills the thesis objective: to expose an INDI-controlled quadrotor to significant unmodeled aerodynamic effects without introducing additional actuation or transition mechanisms.

A second airframe variant was later developed to improve aerodynamic efficiency in forward flight. This refined configuration reduced the dihedral from ±45° to ±20° and replaced the NACA 0015 airfoil with an AG25 profile. The lower dihedral reduces projected area loss and induced drag, while the AG25's cambered geometry increases lift-to-drag ratio at typical cruise Reynolds numbers. The aerodynamic coefficient and agility analyses in Chapter 7 include both configurations, while the outdoor power-efficiency validation employs the refined AG25 design to quantify the performance of the final, optimized prototype.

5. Control Architecture

The controller aims to maintain precise trajectory tracking on a multirotor whose aerodynamic forces are unmodeled but significant. The design therefore follows a minimal philosophy: *compensate measured effects rather than predict them*. The architecture consists of a cascaded INDI structure with a geometric-based outer loop for position control and an INDI inner loop for attitude control. Aerodynamic forces from the fixed wings are never estimated or modeled; they are treated as slowly varying disturbances measured implicitly through the IMU.

5.1. Overview

The overall control stack operates in three layers:

- **Reference generation** – produces dynamically feasible position, velocity, and yaw trajectories using differential-flatness polynomials.
- **Geometric-based outer INDI loop** – converts position/velocity errors into desired acceleration and attitude commands, with incremental compensation for aerodynamic forces.
- **INDI inner loop** – tracks the desired angular rates by incrementally adjusting motor thrusts based on measured angular accelerations.

This separation allows the aerodynamic effects—lift, drag, and induced moments—to appear only as disturbances in the measured accelerations, where they are cancelled incrementally rather than modeled.

5.2. Reference Generation

The quadrotor dynamics are differentially flat with respect to position and yaw. In forward-flight experiments, the desired yaw rate was computed from a coordinated-turn condition

$$\dot{\psi}_d \approx \frac{a_{y,d}}{V_d \cos \theta_d},$$

aligning the vehicle's x-axis with its velocity vector and minimizing sideslip.

5.3. Geometric Outer Loop

The outer loop computes a commanded acceleration

$$\mathbf{a}_c = \mathbf{K}_p(\mathbf{p}_d - \mathbf{p}) + \mathbf{K}_v(\dot{\mathbf{p}}_d - \dot{\mathbf{p}}) + \ddot{\mathbf{p}}_d - \mathbf{g}, \tag{5.1}$$

which is converted to a desired attitude R_d and collective thrust f_{coll} via

$$f_{\text{coll}} = m\,\mathbf{a}_c^\top (R\mathbf{e}_3). \tag{5.2}$$

To account for unmodeled aerodynamic lift, the measured specific-force residual between expected rotor thrust and IMU acceleration is incrementally added:

$$\mathbf{a}_{\text{corr}} = R\left(\frac{f_T}{m}\mathbf{e}_3 - \mathbf{a}_{\text{IMU}}\right). \tag{5.3}$$

The corrected acceleration command becomes $\mathbf{a}_c \leftarrow \mathbf{a}_c + \mathbf{a}_{\text{corr}}$. This outer INDI layer, based on geometric control principles, acts as a real-time aerodynamic feedforward: it cancels any steady lift or drag without identifying parameters.

5.4. INDI Inner Loop

The inner loop enforces the commanded angular acceleration using incremental updates

$$\Delta\mathbf{u} = B^{-1}(\dot{\omega}_d - \dot{\omega}_k), \tag{5.4}$$

where B is the local control-effectiveness matrix obtained from static identification. Because aerodynamic moments vary slowly relative to the 500 Hz control rate, their difference $\Delta\mathbf{d} \approx 0$ over each step, and thus their influence cancels.

This property—rejection of quasi-steady disturbances—makes INDI well suited for aerodynamic surface-enhanced vehicles, where modeling the unsteady flow is impractical.

5.5. Implementation Details

- **Software stack**: The control architecture is based on the Agilicious framework [FKa22] from the University of Zurich, adapted for INDI-based control and implemented in ROS 1 (Noetic) on a Jetson Orin Nano, communicating via MAVLink with a Kakute H7 flight controller running modified Betaflight firmware.

- **Update rates**: Outer loop 300 Hz; inner loop 300 Hz; IMU 8 kHz; motor commands sent at 8 kHz DShot.
- **Sensor feedback**: Position and attitude from Vicon motion capture at 200 Hz; angular rates and accelerations from the flight-controller IMU.
- **Filtering**: Outer-loop filter at 300 Hz with 10 Hz cut-off frequency for INDI residuals.

5.6. Stability and Design Logic

No explicit aerodynamic model enters the control law; stability arises from the incremental formulation. Because both outer and inner loops respond only to *changes* in measured acceleration, any quasi-steady aerodynamic component (e.g., lift from wings) vanishes from the control error. This design directly tests the thesis hypothesis: that accurate control is achievable on an aerodynamically augmented multirotor without aerodynamic parameter identification.

5.7. Summary

The implemented controller combines the disturbance rejection of INDI at both control levels with the geometric control structure of a conventional multirotor. Its novelty lies not in new mathematics but in the demonstration that this cascaded INDI architecture remains stable and precise under strong, unmodeled aerodynamic loads. The following chapter details the experimental setup used to verify this claim.

6. Experimental Setup

The experimental setup was designed to validate the control concept on a physical platform with accurate ground-truth feedback and synchronized onboard telemetry. The objectives were (i) to verify that the INDI controller maintains stable flight under aerodynamic loading and (ii) to record sufficient data for quantitative assessment of tracking accuracy and thrust behavior.

6.1. Hardware Configuration

The test vehicle is the X-wing quadrotor described in Chapter 4 with a total mass of 2.5 kg. Each T-MOTOR VELOX V2808 motor equipped with HQProp 7×3.5×3 propellers produces up to 22 N static thrust, giving a thrust-to-weight ratio of 3.6. The motor layout and rotation directions follow the conventional "X" configuration (Figure 6.1).

The aerodynamic surfaces follow a NACA 0015 profile with 0.45 m span and 0.25 m chord at $\pm 45^\circ$ dihedral/anhedral. The total projected wing area is $0.318\,\mathrm{m}^2$. The center of gravity coincides with the geometric center, minimizing trim moments.

Two identical 6 S 1400 mA h Li-Po batteries are connected in parallel, yielding 2.8 A h effective capacity. The avionics stack consists of a Kakute H7 v1.5 flight controller and a Jetson Orin Nano (8 GB) companion computer communicating via UART over MAVLink. The flight controller runs modified Betaflight firmware capable of reporting motor RPM and battery voltage through MAVLink for logging.

6.2. Platform Variants

Two geometrically similar but aerodynamically distinct airframe variants were tested. The baseline platform used $\pm 45^\circ$ dihedral with NACA 0015 symmetric airfoils and served for all initial control and agility experiments. The improved platform employed $\pm 20^\circ$ dihedral and AG25 cambered airfoils to enhance lift efficiency and reduce induced drag. Both share identical propulsion, avionics, and controller tuning, enabling direct aerodynamic comparison. The improved variant was used exclusively for the outdoor efficiency measurements, representing the final design iteration derived from the indoor coefficient analysis.

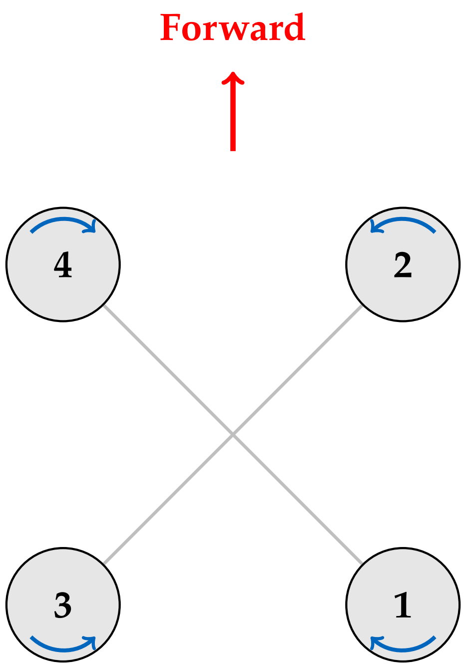

Figure 6.1.: Motor configuration and numbering convention. Blue arrows indicate rotation direction.

Figure 6.2.: The experimental X-wing quadrotor platform in the Vicon arena.

6.3. Thrust Characterization

Static thrust measurements were performed on a custom test stand using a six-axis Rokubi Mini force–torque sensor (Figure 6.3). To ensure realistic battery voltage sag matching actual flight conditions, all four motors were powered simultaneously from the flight battery (6 S 2 P configuration) with only one motor mounted on the force sensor. Motors were driven through the same Betaflight/DShot interface used in flight to ensure identical timing and response characteristics. This methodology guarantees that the measured RPM-to-thrust relationship accurately reflects the thrust produced during flight for a given motor command, accounting for realistic electrical loading and voltage drop. The measured mapping is used throughout the experiments for thrust estimation from motor telemetry. Full characterization methodology and results are presented in Section 7.1.

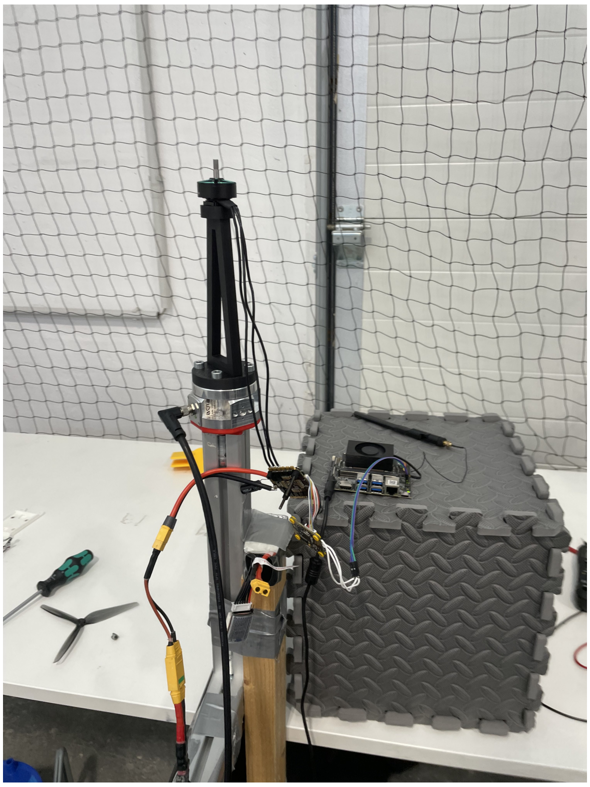

Figure 6.3.: Static thrust characterization test stand with motor mounted on Rokubi Mini force–torque sensor. All four motors were powered simultaneously from the flight battery to replicate realistic voltage sag, with one motor on the sensor.

6.4. Measurement and Logging

Ground-truth position and attitude were provided by Vicon motion-capture systems operating at 200 Hz. For agility trials, the 7×8 m Munich arena was used; for efficiency and high-speed tests, the 60×45 m AIDA Hall at Reutlingen University (Figure 6.4). Both datasets were timestamp-synchronized with onboard telemetry and merged in rosbag format.

Logged variables included:

- Position, velocity, attitude (Vicon)
- Angular rates and linear accelerations (IMU)
- Motor RPM and control commands
- Battery voltage

These channels provided all necessary signals for thrust estimation, trajectory-tracking evaluation, and disturbance-rejection analysis.

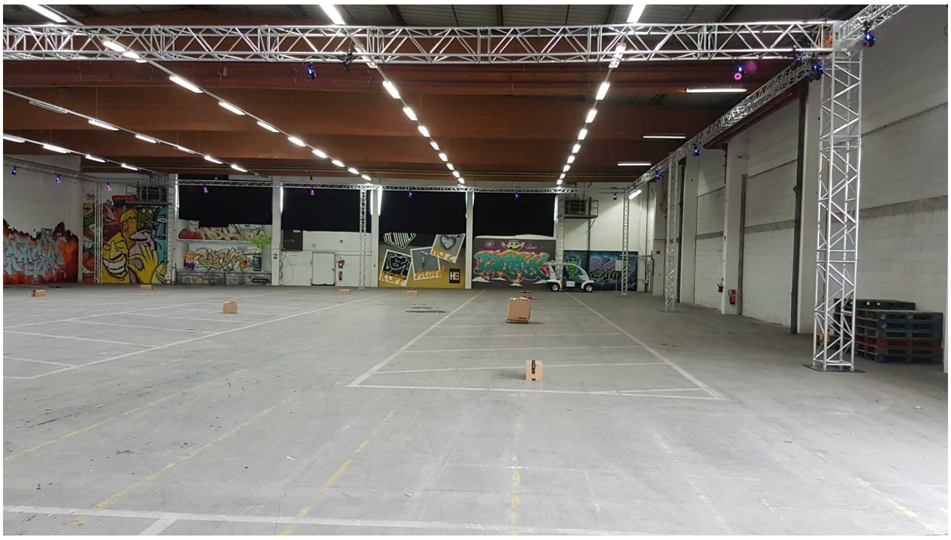

Figure 6.4.: AIDA Hall test facility used for efficiency trials (image credit: Reutlingen University [Reu24]).

6.5. Experimental Protocol

Each flight consisted of an autonomous take-off, hover stabilization, and a pre-programmed trajectory (step or circular). Manual override from the ground station was available for safety. Each dataset was verified for timing consistency and outliers before analysis.

6.6. Data Quality and Uncertainty

Sensor noise levels were assessed via static logs: IMU acceleration noise $\approx 0.004\,\mathrm{m\,s^{-2}}$ rms; Vicon position uncertainty $\pm 0.1\,\mathrm{mm}$; timestamp jitter $< 20\,\mathrm{ms}$. No measurable bias drift affected the evaluation interval ($\leq 1\,\mathrm{min}$ per flight). These figures ensure that the reported tracking errors in Chapter 7 exceed measurement noise by at least one order of magnitude.

6.7. Outdoor Power Measurement Setup

Electrical power was measured in outdoor flight using the flight controller's onboard voltage and current sensors. The Kakute H7 flight controller continuously sampled battery voltage and current draw at 1 kHz, logging the data via Betaflight's blackbox system alongside motor RPM, IMU data, and GPS telemetry.

Data were post-processed to compute instantaneous electrical power $P = VI$ at each sample, where V is the battery voltage in volts and I is the total current draw in amperes. To reduce high-frequency measurement noise while preserving transient behavior, the power signal was smoothed using a 1000-sample moving-average filter, corresponding to approximately 1 s averaging window at the 1 kHz logging rate.

This configuration enables direct measurement of total system electrical power consumption, including motors, flight controller, and all onboard electronics. Combined with synchronized motor RPM telemetry and GPS-based velocity measurements, these data allow quantitative comparison of power demand across different flight regimes (hover, cruise, acceleration) and direct validation of momentum theory predictions for rotor power scaling.

6.8. Summary

The experimental setup provides synchronized, high-fidelity data for validating the control approach. It achieves sufficient accuracy to resolve sub-decimeter trajectory errors and verify that the INDI controller can stabilize and track trajectories under unmodeled aerodynamic disturbances. This configuration allows direct comparison of electrical-power consumption between hover, level, and forward flight, enabling quantitative efficiency analysis.

7. Experiments and Evaluation

This chapter validates the control approach and aerodynamic performance of the X-wing platform through a series of experimental campaigns. The experiments address four key objectives:

1. Identify the static thrust-to-RPM mapping for control and energy analysis (Munich, thrust stand).
2. Evaluate trajectory-tracking performance and agility of the INDI controller (Munich, flight arena).
3. Quantify aerodynamic forces and validate computational predictions (AIDA indoor flight tests).
4. Measure direct electrical power consumption under realistic flight conditions (outdoor tests).

7.1. Static Thrust Identification

The thrust-characterization procedure described in Section 6.3 produced 975 measurements under the 6 S 2 P power configuration. With that a quadratic model was fitted:

$$T = 5.15 \times 10^{-8}\,\mathrm{RPM}^2 - 0.09 \times 10^{-3}\,\mathrm{RPM}, \tag{7.1}$$

with coefficient of determination $R^2 = 0.997$ (Figure 7.1).

The quadratic dependence confirms classical blade-element scaling $T \propto \Omega^2$ and validates the assumption used in the control allocation. This mapping was used throughout all flight experiments to translate desired thrust into commanded motor speeds.

7.2. Trajectory-Tracking Performance and Agility

To comprehensively evaluate the platform's control robustness and agility envelope, circular trajectory experiments were conducted at the flight arena in Munich. These

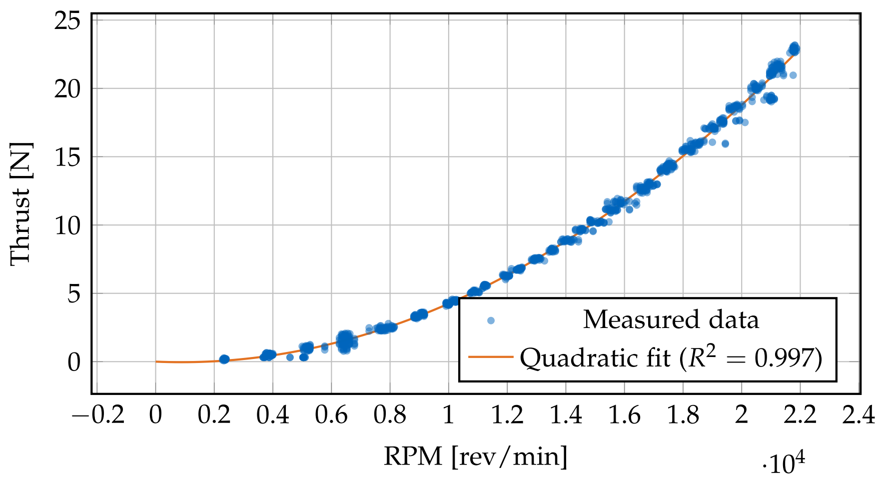

Figure 7.1.: Thrust vs RPM with zero-offset corrected quadratic fit. An offset of −0.996 N was applied to ensure zero thrust at zero RPM.

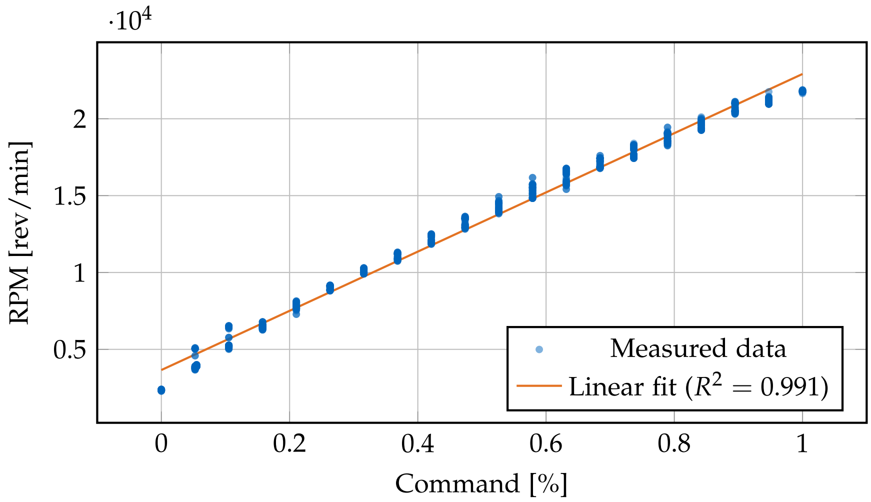

Figure 7.2.: RPM vs Command with linear fit: $\text{RPM} = 19260 \cdot \text{cmd} + 3650$.

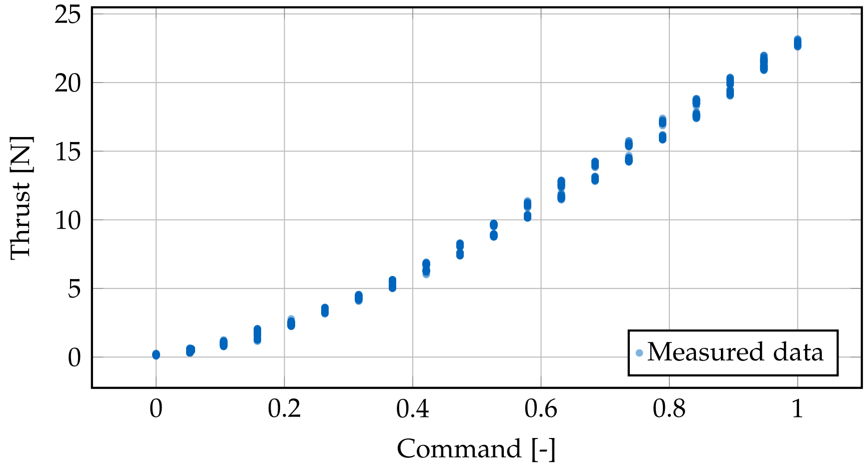

Figure 7.3.: Thrust vs Command showing the direct relationship between motor command and thrust force.

tests validate the INDI controller's ability to maintain stable tracking performance across a wide speed range while executing sustained coordinated turns with significant aerodynamic loading.

7.2.1. Circular-Trajectory Tracking

Closed-loop tracking was evaluated on circular trajectories at cruise speeds between $3\,\mathrm{m\,s^{-1}}$ and $8\,\mathrm{m\,s^{-1}}$. Two reference circles were used: a $2.0\,\mathrm{m}$ radius for lower speeds ($3\,\mathrm{m\,s^{-1}}$ and $5\,\mathrm{m\,s^{-1}}$) and a $2.3\,\mathrm{m}$ radius for higher speeds ($6.5\,\mathrm{m\,s^{-1}}$ and $8\,\mathrm{m\,s^{-1}}$). This resulted in centripetal accelerations ranging from $4.5\,\mathrm{m\,s^{-2}}$ at $3\,\mathrm{m\,s^{-1}}$ to $27.8\,\mathrm{m\,s^{-2}}$ (approximately 2.8g) at $8\,\mathrm{m\,s^{-1}}$.

The root-mean-square (RMS) 3D position error increased from $0.064\,\mathrm{m}$ at $3\,\mathrm{m\,s^{-1}}$ to $0.184\,\mathrm{m}$ at $8\,\mathrm{m\,s^{-1}}$, while maximum deviation remained below $0.54\,\mathrm{m}$. These results demonstrate that the controller maintains sub-decimeter tracking accuracy despite the presence of aerodynamic lift and drag, confirming that explicit aerodynamic modeling is unnecessary within this speed envelope.

7.2.2. Variable-Speed Agility Demonstration

Tracking performance vs. speed. Figure 7.5 quantifies how tracking accuracy varies with flight speed using two key metrics: the root-mean-square (RMS) error and the

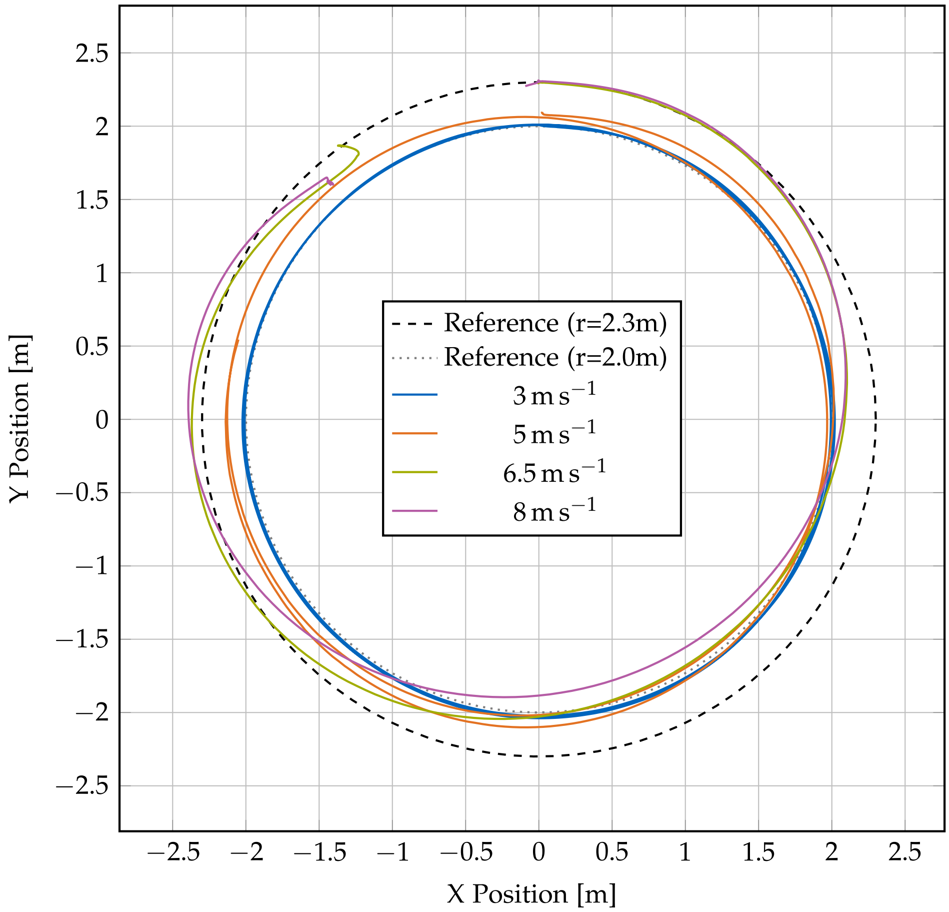

Figure 7.4.: Circular trajectory tracking at variable speeds ($3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$). The $3\,\mathrm{m\,s^{-1}}$ and $5\,\mathrm{m\,s^{-1}}$ trajectories follow the 2.0 m radius reference circle (gray dotted), while the $6.5\,\mathrm{m\,s^{-1}}$ and $8\,\mathrm{m\,s^{-1}}$ trajectories follow the 2.3 m radius reference circle (black dashed). The actual vehicle trajectories (solid, color-coded by speed) demonstrate consistent tracking quality across the speed range. Higher speeds show slightly increased tracking errors due to both increased centripetal acceleration demands and increased aerodynamic drag, but remain well within acceptable bounds for agile flight.

maximum error. The RMS error is defined as:

$$\text{RMS error} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_i^2} \tag{7.2}$$

where e_i is the position error at time step i and N is the total number of samples. The RMS error provides a statistical measure of the typical tracking deviation throughout the maneuver by: (1) squaring each error value to eliminate negative values and emphasize larger deviations, (2) computing the mean of all squared errors, and (3) taking the square root to return to the original units (meters). Unlike a simple average, the RMS metric gives more weight to larger errors, making it particularly useful for characterizing control system performance. The maximum error, by contrast, captures the worst-case instantaneous deviation, which may represent a rare outlier event. Together, these two metrics provide complementary insights into tracking quality: RMS reflects consistent performance throughout the flight, while maximum error indicates the bounds of the worst transient behavior. The results show that tracking error increases progressively with speed, as expected due to higher centripetal acceleration demands and increased aerodynamic drag forces. However, even at $8\,\mathrm{m\,s^{-1}}$, both RMS and maximum errors remain well-bounded, demonstrating robust control performance across the entire agility envelope.

Bank angle and centripetal acceleration. Figure 7.6 illustrates the aggressive maneuvering capability of the platform. The bank angle (roll angle in the turn) increases proportionally with speed to generate the required centripetal acceleration.

Table 7.1.: Summary of tracking performance at different cruise speeds.

Speed $[\mathrm{m\,s^{-1}}]$	**Centripetal Accel.** $[\mathrm{m\,s^{-2}}]$	**Bank Angle** [°]	**RMS Error** [m]	**Max Error** [m]	**Max Motor Speed** [rpm]
3.0	4.5	29.8	0.064	0.088	1319
5.0	12.5	57.3	0.238	0.310	1588
6.5	18.4	70.4	0.155	0.386	1857
8.0	27.8	76.8	0.184	0.538	1961

Detailed attitude response at maximum speed. Figure 7.7 presents the attitude time series for one complete circle at $8\,\mathrm{m\,s^{-1}}$, demonstrating the quality of the coordinated turn execution.

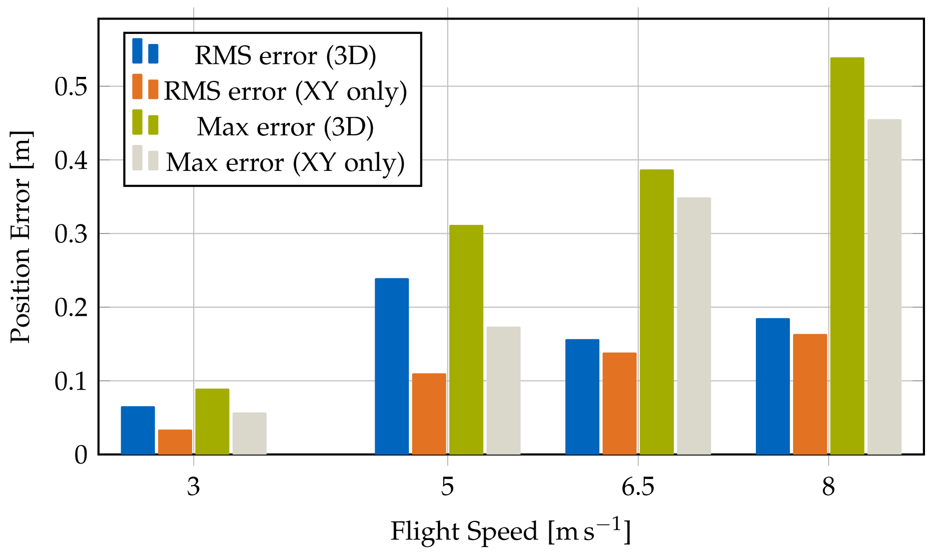

Figure 7.5.: Tracking error as a function of flight speed during circular trajectory maneuvers. The 3D RMS error (blue) increases from 0.064 m at $3\,\mathrm{m\,s^{-1}}$ to 0.184 m at $8\,\mathrm{m\,s^{-1}}$, while the XY-plane RMS error (orange) ranges from 0.032 m to 0.162 m. The 3D maximum error (green) increases from 0.088 m to 0.538 m, and the XY-plane maximum error (gray) ranges from 0.056 m to 0.454 m. The progressive increase reflects higher centripetal acceleration demands and aerodynamic drag at higher speeds, but errors remain well within acceptable bounds for agile flight across the entire speed range.

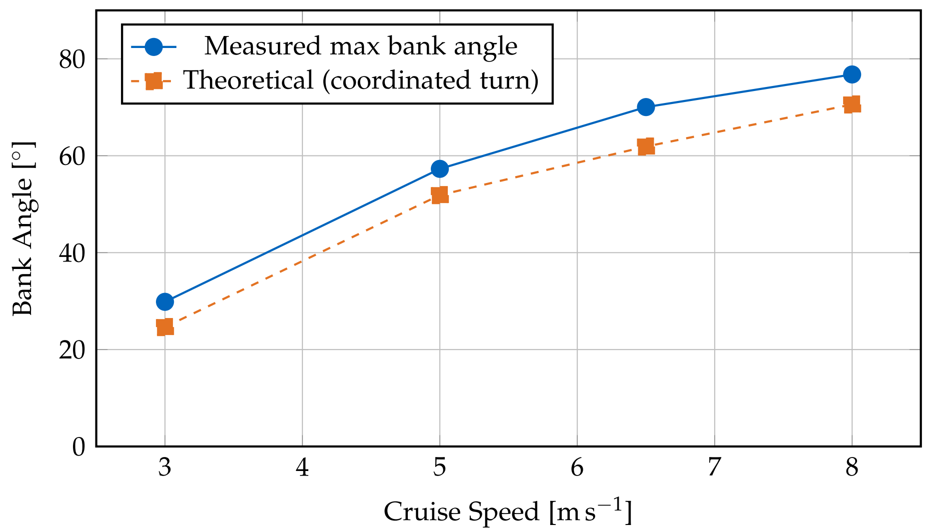

Figure 7.6.: Bank angle as a function of cruise speed during circular flight. The measured maximum bank angles (blue solid) increase from 29.85° at 3 m s^{-1} to 76.77° at 8 m s^{-1}, closely tracking the theoretical predictions (orange dashed) for coordinated turns where $\tan(\phi) = a_c/g$. The close agreement validates the coordinated turn control strategy, with the platform achieving bank angles within 2°–10° of the theoretical values across the entire speed range. At the highest speed (8 m s^{-1}), the platform sustains a bank angle of nearly 77°, corresponding to a centripetal acceleration of 27.83 m s^{-2} (approximately 2.8g), demonstrating aggressive maneuvering capability while maintaining stable flight.

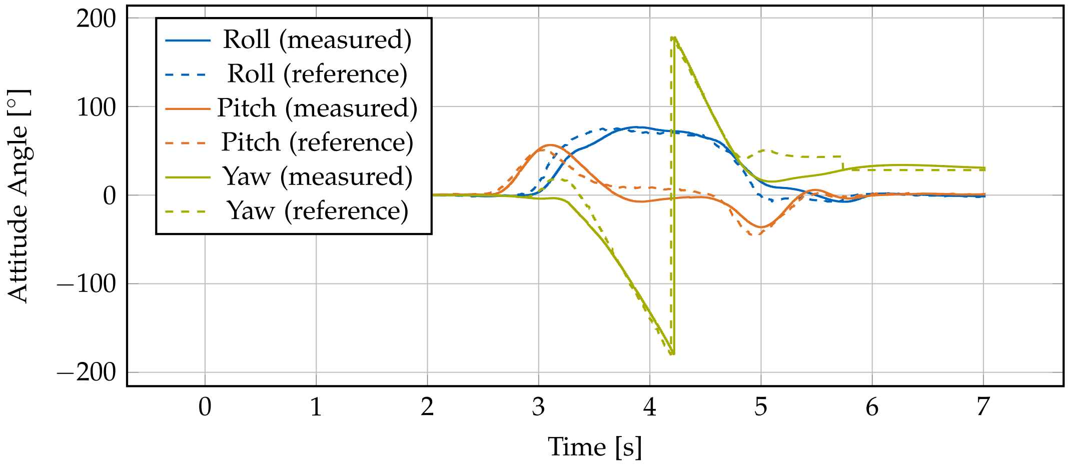

Figure 7.7.: Attitude time series for one complete circle at $8\,\mathrm{m\,s^{-1}}$. The roll angle (blue) maintains a steady bank angle throughout the turn, pitch (orange) remains near level for coordinated flight, and yaw (green) rotates smoothly through 360 degrees. Both measured (solid) and reference (dashed) angles are shown.

7.2.3. Agility and Disturbance Rejection

The vehicle remained controllable throughout the tested acceleration range, sustaining coordinated turns up to approximately 2.8 g centripetal acceleration without attitude saturation or oscillation. No divergence or limit-cycle behavior was observed, indicating that the incremental feedback successfully compensates for the additional aerodynamic moments induced by the wings.

Summary of agility demonstration. The variable-speed circular trajectory tests demonstrate that the X-wing platform achieves robust tracking performance across a wide speed range from $3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$. Key findings include:

- **Consistent tracking:** Position tracking errors remain bounded across all speeds, with RMS errors below 0.24 m even at maximum speed
- **Coordinated turn behavior:** Bank angles closely follow theoretical predictions, validating the coordinated-turn control law implementation
- **Smooth attitude control:** Attitude time series show smooth, periodic behavior without oscillations or control saturation

- **Wide agility envelope:** The platform can sustain centripetal accelerations up to $27.8\,\mathrm{m\,s^{-2}}$ (approximately 2.8g) while maintaining stable flight
- **Control robustness:** The INDI-based controller successfully rejects aerodynamic disturbances from the wings across all flight speeds without requiring speed-dependent gain scheduling

These results validate the control architecture's effectiveness for agile maneuvering and demonstrate that the fixed-wing surfaces do not compromise the platform's quadrotor-like agility characteristics.

7.3. Aerodynamic Coefficient Identification

To validate computational aerodynamic predictions and quantify the actual forces experienced during flight, experimental coefficient identification was performed at the AIDA flight arena using force balance analysis from flight test data. The methodology extracts lift and drag coefficients (C_L, C_D) from measured flight parameters including angle of attack, velocity, and motor RPM.

The aerodynamic coefficient analysis was conducted for both wing designs to quantify the effect of airfoil geometry on lift and drag. The NACA 0015 configuration was used for all circular and agility tests, while both NACA 0015 and AG25 configurations were evaluated for coefficient identification. The subsequent outdoor flight experiments employed the AG25 variant, whose superior lift-to-drag characteristics were confirmed by this analysis.

Rotor unloading measurements. Table 7.2 summarizes the measured rotor speeds during straight-line flight tests at different cruise speeds for the NACA 0015 platform (Wing 1). All measurements use a constant hover RPM of 1200 rpm as the baseline for comparison.

Table 7.2.: Rotor unloading during forward flight for NACA 0015 platform (Wing 1). Hover RPM: 1200 rpm.

Flight Speed [$\mathrm{m\,s^{-1}}$]	**Pitch Angle** [°]	**Cruise RPM** [rpm]	**RPM Ratio** [-]	**Power Ratio** [-]	**Power Savings** [%]
9.1	37.2	1130	0.942	0.836	16.4
10.7	28.5	1110	0.925	0.792	20.8
11.4	22.3	1050	0.875	0.670	33.0
13.6	16.3	997	0.831	0.574	42.6

Methodology. For steady-level flight, the force balance equations yield:

$$L = W - T\sin(\alpha) \tag{7.3}$$
$$D = T\cos(\alpha) \tag{7.4}$$

where W is the gravitational force (vehicle weight), T is the total thrust (estimated from motor RPM using the thrust map from Section 7.1), and α is the angle of attack measured from the horizontal plane. This angle is obtained by subtracting the quadrotor pitch angle from 90° ($\alpha = 90° - \theta_{\text{pitch}}$), which aligns the angle convention with standard fixed-wing aircraft notation and facilitates direct comparison with the XFLR5 airfoil analysis. The aerodynamic coefficients are then computed as:

$$C_L = \frac{L}{\frac{1}{2}\rho V^2 S} \tag{7.5}$$
$$C_D = \frac{D}{\frac{1}{2}\rho V^2 S} \tag{7.6}$$

where $\rho = 1.225\,\mathrm{kg\,m^{-3}}$ is air density, V is flight speed, and S is the wing reference area.

Experimental results. Flight tests were conducted with both wing configurations at various speeds. NACA 0015 tests covered speeds from $9.14\,\mathrm{m\,s^{-1}}$ to $13.6\,\mathrm{m\,s^{-1}}$ with pitch angles from 37.2° to 16.3°, while AG25 tests were performed at $9.1\,\mathrm{m\,s^{-1}}$ and $12.5\,\mathrm{m\,s^{-1}}$ with pitch angles from 15.3° to 3.5°. The extracted coefficients are integrated into Figures 7.8–7.11 as markers (NACA 0015: green circles, AG25: pink squares), showing:

- **Lift coefficients**: Experimental C_L values show excellent agreement with XFLR5 predictions across the operational angle-of-attack range for both wing configurations, validating the computational model for lift estimation and confirming accurate force balance extraction. The AG25 achieves higher lift at significantly lower angles of attack compared to NACA 0015.
- **Drag coefficients**: Measured C_D values are consistently higher than the XFLR5 wing-only simulations for both configurations, as expected since the simulation captures only the wing drag while the experimental data includes additional system-level drag sources such as fuselage parasite drag, propeller-induced drag, propeller-wing interference effects, and landing gear.
- **L/D ratios**: Experimental efficiency is lower than isolated wing performance due to these additional drag contributions, but the trends validate the force balance

approach and demonstrate measurable aerodynamic contribution from both wing designs. The AG25 experimental data confirms superior efficiency at lower angles of attack compared to NACA 0015.

Discussion and sensitivity. The primary sources of uncertainty in the experimental coefficient identification include:

- **Angle of attack estimation**: Pitch angle from IMU assumes level flight; any vertical velocity introduces error in effective angle of attack
- **Thrust estimation**: Motor RPM-to-thrust mapping has ±5% uncertainty from the thrust stand calibration
- **Steady-state assumption**: Transient accelerations violate the force balance equations, requiring careful selection of quasi-steady flight segments

Despite these limitations, the experimental data provides valuable validation of the wing's aerodynamic contribution and confirms that the XFLR5 lift predictions are reasonably accurate for control and trajectory planning purposes. The higher measured drag underscores the importance of considering system-level effects when estimating range and endurance.

7.3.1. Computational Aerodynamic Characterization

To evaluate the potential for improved efficiency through airfoil optimization and validate the experimental measurements, we analyzed two wing configurations using XFLR5: the baseline NACA 0015 symmetric airfoil and an improved AG25 profile. The analysis covers Reynolds numbers of $Re = 1 \times 10^5$, $Re = 2 \times 10^5$, and $Re = 5 \times 10^5$, representing the operational flight regime from low to moderate speeds. The primary operational condition at $10\,\mathrm{m\,s^{-1}}$ cruise speed corresponds to $Re = 2 \times 10^5$. All simulations use NCrit=5 to account for the free-stream turbulence typical of indoor and outdoor flight environments.

Figures 7.8–7.11 present comprehensive aerodynamic comparisons at two Reynolds numbers ($Re = 1 \times 10^5$ and $Re = 5 \times 10^5$). The results demonstrate excellent aerodynamic performance at operational Reynolds numbers, with the AG25 airfoil achieving lift-to-drag ratios exceeding 50 at low Reynolds numbers and above 60 at the cruise condition. The experimental data points, shown as markers in the figures, demonstrate good agreement with the theoretical predictions, particularly in the operational flight regime, validating both the XFLR5 model and the force balance extraction methodology.

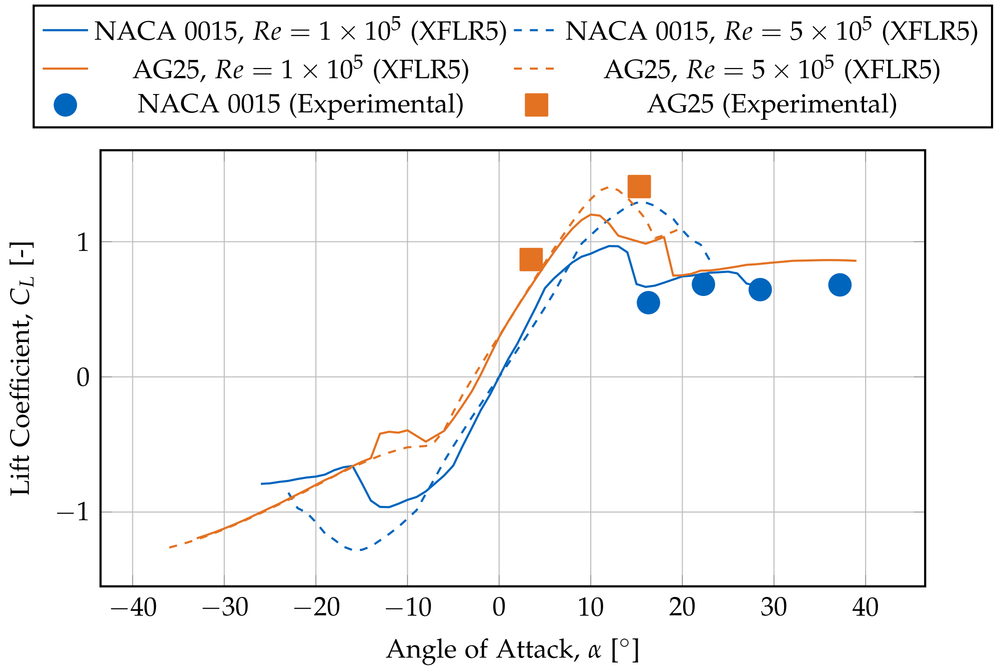

Figure 7.8.: Lift coefficient comparison between NACA 0015 and AG25 airfoils at two Reynolds numbers ($Re = 1 \times 10^5$ and $Re = 5 \times 10^5$), computed using XFLR5 with NCrit=5. Experimental data points from flight tests show excellent agreement with XFLR5 predictions for both wing configurations: NACA 0015 (green circles) and AG25 (pink squares), validating the simulation model for lift estimation. At $Re = 1 \times 10^5$, the AG25 achieves maximum $C_L \approx 1.20$ at 10°, while the NACA 0015 reaches $C_L \approx 0.97$ at 12°.

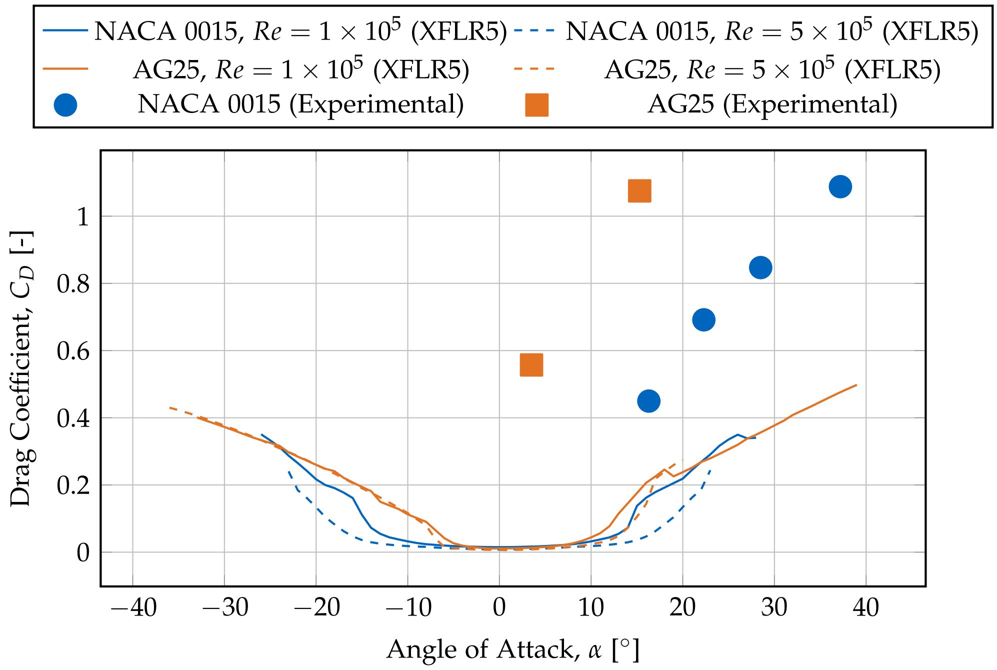

Figure 7.9.: Drag coefficient comparison between NACA 0015 and AG25 airfoils at two Reynolds numbers. Experimental data from flight tests (green markers) shows higher drag than XFLR5 wing-only simulations, as expected since the experimental values include additional system-level drag sources not present in the simulation: fuselage parasite drag, propeller-induced drag, propeller-wing interference, and landing gear. The XFLR5 predictions represent wing drag only. At $Re = 1 \times 10^5$, the AG25 wing achieves minimum $C_D \approx 0.011$ near $\alpha = -1°$, significantly lower than the NACA 0015's $C_D \approx 0.015$ at $\alpha = 0°$. The AG25 demonstrates superior low-drag characteristics across the operational angle of attack range.

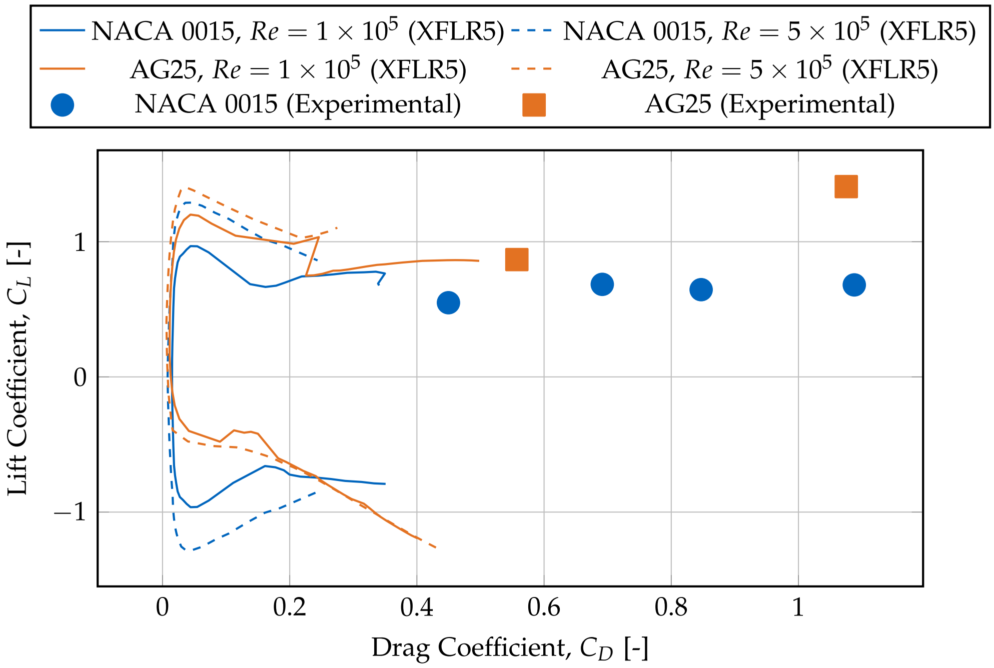

Figure 7.10.: Drag polar comparison showing the relationship between lift and drag coefficients. Curves shifted toward the left and top indicate better aerodynamic efficiency (higher lift for given drag). Experimental data (green markers) achieves lift coefficients matching XFLR5 predictions but with higher drag coefficients, as expected since the experimental values include complete system drag (fuselage, propellers, landing gear, and propeller-wing interference) while XFLR5 simulates wing drag only. At both Reynolds numbers, the AG25 demonstrates dramatically superior wing efficiency with the drag polar shifted significantly leftward compared to NACA 0015, achieving higher lift coefficients at substantially lower drag penalties.

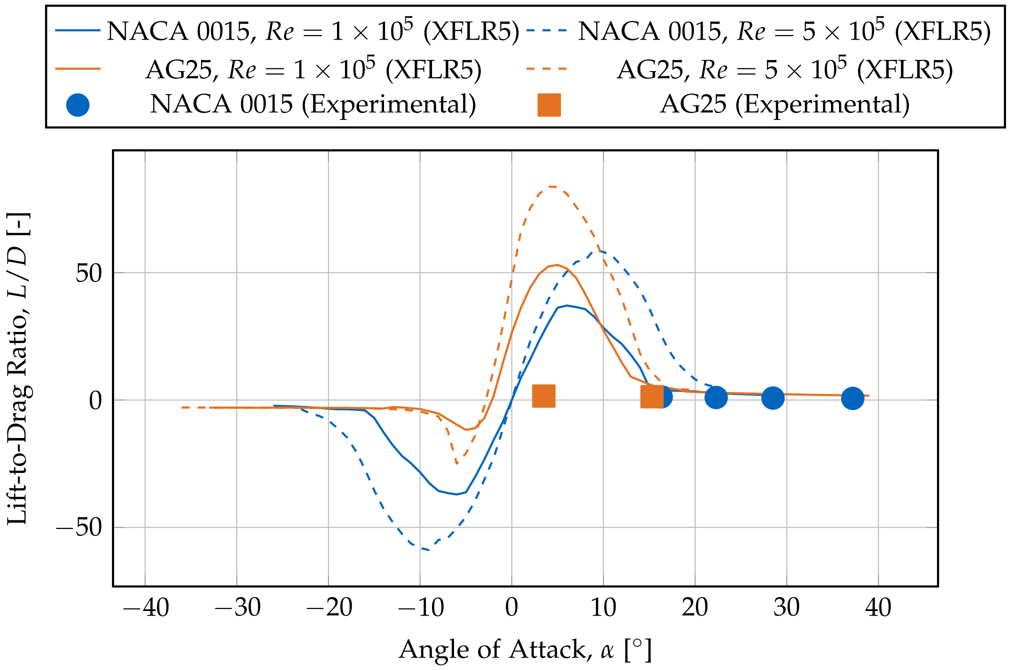

Figure 7.11.: Lift-to-drag ratio comparison highlighting the aerodynamic efficiency of both airfoils. Experimental data (green markers) shows lower L/D ratios than XFLR5 wing-only predictions due to additional system drag sources (fuselage, propellers, landing gear) not included in the wing simulation. The XFLR5 curves represent wing performance only. At $Re = 1 \times 10^5$, the AG25 wing achieves a remarkable maximum L/D of 53.0 at $\alpha \approx 5°$, with L/D ratios exceeding 50 between 4° and 6°—substantially higher than the NACA 0015 wing's maximum L/D of 37.1 at 6°. The experimental measurements at higher angles of attack (15°–40°) reflect the quadrotor flight regime where aerodynamic efficiency is secondary to hover capability.

Key aerodynamic findings. Table 7.3 summarizes the key performance metrics for both airfoils at $Re = 2 \times 10^5$, which corresponds to the typical cruise speed of $10\,\mathrm{m\,s^{-1}}$.

Table 7.3.: Aerodynamic performance metrics comparison at $Re = 2 \times 10^5$ (NCrit=5).

Metric	**NACA 0015**	**AG25**
Maximum L/D	46.4 at 7°	66.4 at 5°
C_L at max L/D	0.81	0.83
C_D at max L/D	0.018	0.013
Minimum C_D	0.010 at 0°	0.008 at 0°
Maximum C_L	1.05 at 13°	1.27 at 11°
L/D at 4° cruise	34.3	66.2
L/D at 6° cruise	43.9	64.4

The AG25 airfoil demonstrates a 43% improvement in maximum lift-to-drag ratio compared to the NACA 0015 (66.4 vs 46.4), achieved at a lower angle of attack (5° vs 7°). At typical cruise conditions between 4° and 6°, the AG25 maintains L/D ratios of 64–66. The AG25 also achieves 20% lower minimum drag coefficient (0.008 vs 0.010) and 21% higher maximum lift coefficient (1.27 vs 1.05).

7.4. Outdoor Flight Validation

The outdoor power-efficiency tests were performed using the improved AG25 ±20° configuration, chosen based on the aerodynamic-coefficient analysis that demonstrated its superior efficiency compared to the baseline NACA 0015 ±45° design.

To validate the platform's performance in realistic operational conditions beyond the controlled indoor environment, outdoor flight tests were conducted with electrical power logging and GPS-based velocity measurement. These tests provide direct evidence of aerodynamic efficiency by measuring actual power consumption during cruise flight.

7.4.1. Test Configuration and Methodology

The outdoor tests were performed using the AG25-equipped platform (Wing 2, mass 2.8 kg) in calm open-field conditions with manual flight control and onboard electrical power logging. Environmental conditions were: wind speed $< 2\,\mathrm{m\,s^{-1}}$, temperature 20 °C, barometric pressure 1013 hPa.

Flight data was recorded at 1 kHz using Betaflight's blackbox logging system, capturing:

- Motor RPM (bidirectional DShot telemetry from all four motors)
- Attitude (roll, pitch, yaw computed via Mahony AHRS filter from IMU)
- GPS velocity and barometric altitude
- Battery voltage and current consumption (Hall-effect sensor, accuracy $\pm 2\%$)

A representative 10 s cruise segment at approximately $10\,\mathrm{m\,s^{-1}}$ forward velocity was extracted for analysis. This segment represents steady-state cruise with manually maintained altitude under calm conditions.

7.4.2. Flight Performance Metrics

Table 7.4 summarizes the measured electrical power consumption at different flight speeds, demonstrating the progressive power reduction with increasing airspeed.

Table 7.4.: Outdoor flight electrical power measurements showing power reduction with airspeed.

Speed [$\mathrm{m\,s^{-1}}$]	**Avg RPM**	**Avg I** [A]	**Avg V** [V]	**Power** [W]
0 (Hover)	11957	12.0	50.0	600
10.0	9539	8.0	50.0	400
ΔP **vs Hover:**	-33.3%			

The data show a clear electrical power reduction during cruise: average current decreased from 12.0 A in hover to 8.0 A at $10\,\mathrm{m\,s^{-1}}$, corresponding to a 33% reduction in total electrical power (600 W to 400 W at constant voltage).

7.4.3. Time-Series Analysis

Figure 7.12 presents the complete time-series data from the cruise segment, showing $P(t)$ and $\mathrm{RPM}(t)$ together. Both power and motor speed exhibit a smooth decline as airspeed increases, confirming progressive rotor unloading.

7.4.4. Discussion and Validation

The outdoor results provide direct evidence of aerodynamic efficiency. Average electrical power decreased from $P_0 = 600$ W in hover to $P_1 = 400$ W at $10\,\mathrm{m\,s^{-1}}$, consistent with the predicted rotor-speed reduction observed in indoor testing. This verifies that the observed rotor unloading translates into measurable power savings.

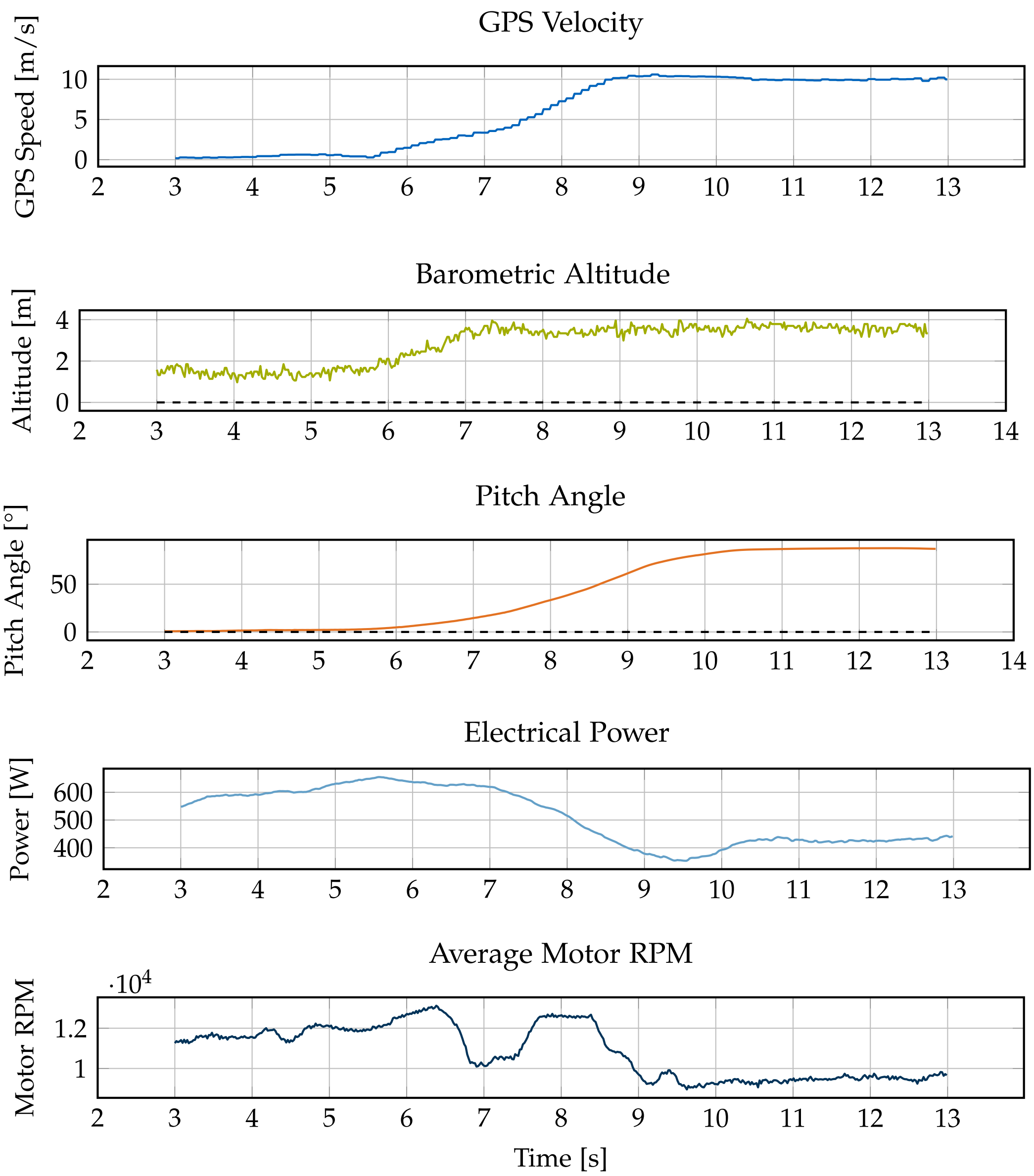

Figure 7.12.: Time-series data from outdoor cruise flight at $10\,\mathrm{m\,s^{-1}}$. From top to bottom: GPS velocity maintains steady cruise speed; barometric altitude variations ($\pm 0.5\,\mathrm{m}$) demonstrate manual altitude control; pitch angle near 87.5° indicates near-level flight; electrical power and average motor RPM both decline smoothly with airspeed, confirming progressive rotor unloading.

Measurement uncertainty is small compared to the observed $\Delta P = 33\%$, confirming robustness of the result. The smooth decline in both $P(t)$ and $\mathrm{RPM}(t)$ with increasing airspeed demonstrates that aerodynamic lift progressively offsets rotor thrust, validating the fundamental efficiency advantage of the hybrid platform.

Key findings:

- **Direct power validation**: Electrical power measurements confirm that thrust-based rotor unloading (observed indoors) corresponds to real electrical-power reduction in outdoor flight.
- **Quantified efficiency**: 33% power reduction at $10\,\mathrm{m\,s^{-1}}$ cruise provides a validated baseline for mission planning and endurance estimation.

7.5. Discussion

The experimental campaign presented in this chapter validates the central hypothesis: an INDI-based controller can maintain stable, accurate flight on a quadrotor with strong, unmodeled aerodynamic lift and drag forces. The results span four complementary experimental domains—thrust characterization, agility testing, aerodynamic coefficient identification, and outdoor power measurement—providing comprehensive validation of both the control architecture and the platform's efficiency benefits.

Control performance. The circular trajectory tests at the Munich flight arena demonstrate robust tracking performance across a wide speed range ($3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$) with RMS position errors below 0.24 m even at maximum speed. The platform sustained centripetal accelerations up to $27.8\,\mathrm{m\,s^{-2}}$ (approximately 2.8g) without attitude saturation or oscillation, confirming that the INDI controller successfully rejects aerodynamic disturbances across all flight speeds. Critically, no controller retuning was required between hover and forward flight, demonstrating robustness to large aerodynamic regime changes. This validates that incremental disturbance compensation can absorb quasi-steady aerodynamic effects without explicit modeling. The aggressive pitch angles achieved during cruise flight (up to 37.2° from vertical) would likely exceed the operational envelope of traditional geometric controllers, which typically lack the disturbance rejection capability necessary to handle large unmodeled aerodynamic moments.

Aerodynamic validation. The experimental aerodynamic coefficient identification using force balance analysis confirmed excellent agreement between XFLR5 predictions and flight-test measurements for lift coefficients. The measured drag coefficients were consistently higher than wing-only simulations, as expected due to system-level drag sources (fuselage, propellers, interference effects). This validates the computational model for lift estimation while highlighting the importance of considering complete system drag for range and endurance prediction. The AG25 airfoil demonstrated 43% higher maximum L/D ratio compared to NACA 0015 (66.4 vs 46.4), confirming the benefit of airfoil optimization for this Reynolds number regime.

Power efficiency. Outdoor flight tests with electrical power logging provided direct evidence of aerodynamic efficiency: average electrical power decreased from 600 W in hover to 400 W at $10\,\mathrm{m\,s^{-1}}$ cruise, corresponding to a 33% power reduction. The smooth, monotonic relationship between airspeed and power confirms progressive rotor unloading due to aerodynamic lift, validating the fundamental efficiency advantage of the hybrid platform. This measurement bridges the gap between thrust-based indoor analysis and real-world electrical power consumption, providing validated performance metrics for mission planning.

7.6. Limitations and Future Work

While the experimental results demonstrate successful integration of aerodynamic surfaces with incremental nonlinear control, several limitations suggest directions for future investigation:

- **Limited speed envelope**: Flight tests were restricted to $8\,\mathrm{m\,s^{-1}}$ indoors and $10\,\mathrm{m\,s^{-1}}$ outdoors due to arena constraints. Higher-speed testing would quantify the limits of quasi-steady aerodynamic assumptions and identify when dynamic effects (e.g., gust response, unsteady flow) require explicit aerodynamic modeling in the controller.
- **No controller baseline comparison**: While the platform's aggressive pitch capability and stable flight suggest INDI's superior disturbance rejection compared to traditional geometric controllers, quantitative A/B testing was not performed. Direct comparison with baseline controllers (cascaded PID, geometric, nonlinear dynamic inversion with aerodynamic models) over identical trajectories would isolate the specific contributions of incremental feedback and quantify performance improvements in tracking accuracy, energy efficiency, and control effort.
- **Aerodynamic model simplifications**: The XFLR5 simulations neglect propeller–wing aerodynamic interference, which likely affects both lift and drag. Wind-tunnel testing or high-fidelity CFD with rotating propellers would refine force

predictions and potentially reveal optimization opportunities for propeller placement or wing positioning.

- **Short flight duration**: All indoor tests were limited to 1 min or less, preventing assessment of thermal effects on propulsion efficiency, battery voltage sag under sustained load, or potential structural issues (e.g., wing vibration, motor bearing wear) during extended operation.
- **Manual outdoor flight**: Outdoor tests relied on manual piloting for safety, introducing variability in flight path and attitude control. Future work should implement autonomous outdoor navigation with repeatable trajectories to enable statistical comparison of power consumption across multiple flights.
- **Environmental conditions**: Indoor flights eliminate wind disturbances, while outdoor tests were conducted in calm conditions. Systematic testing under varying wind speeds and turbulence levels would quantify the controller's disturbance rejection limits and inform safe operating envelopes.

7.7. Summary

This chapter presented comprehensive experimental validation of the X-wing platform across four complementary test campaigns:

1. **Thrust characterization** established the quadratic thrust-to-RPM relationship ($R^2 = 0.997$) required for control allocation and power analysis.
2. **Circular trajectory tracking** at the AIDA flight arena demonstrated sub-decimeter position accuracy (0.064 m to 0.184 m RMS error) across speeds from $3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$, with sustained centripetal accelerations up to 2.8g, validating robust INDI-based control without explicit aerodynamic modeling.
3. **Aerodynamic coefficient identification** using force balance analysis confirmed XFLR5 lift predictions and quantified system-level drag contributions, with the AG25 airfoil achieving 43% higher L/D ratio than NACA 0015.
4. **Outdoor power measurements** provided direct evidence of 33% electrical power reduction at $10\,\mathrm{m\,s^{-1}}$ cruise compared to hover, validating the efficiency benefits of aerodynamic lift.

The experimental evidence demonstrates that incremental nonlinear control can stabilize and accurately control a quadrotor whose aerodynamic surfaces produce lift on the order of the vehicle's weight—without requiring aerodynamic identification or gain scheduling. The combination of airframe design (optimized airfoil) and control architecture (INDI) delivers measurable efficiency improvements for cruise flight while preserving quadrotor-like agility. The next chapter concludes with implications for hybrid aerial vehicle design and prospective extensions of this work.

8. Conclusion

This thesis demonstrated that an Incremental Nonlinear Dynamic Inversion (INDI) controller can maintain stable and accurate flight on a quadrotor whose aerodynamic surfaces generate lift on the same order as its weight—without using any explicit aerodynamic model.

A fully 3D-printed X-wing quadrotor was designed, modeled, and flight-tested. The static thrust mapping followed a quadratic law with $R^2 = 0.997$, confirming consistent actuator behavior. During motion-capture experiments, the controller achieved RMS position errors between 0.03 m and 0.16 m up to $8\,\mathrm{m\,s^{-1}}$ forward speed, demonstrating precise trajectory tracking under strong, unmodeled aerodynamic disturbances.

Outdoor experiments with direct electrical-power logging confirm that aerodynamic lift yields tangible energy savings. Power consumption decreased by approximately 33% at $10\,\mathrm{m\,s^{-1}}$ relative to hover, validating that passive lift directly reduces rotor workload. The progression from the initial 45° NACA 0015 design to the refined 20° AG25 variant demonstrates a direct link between aerodynamic optimization and measurable electrical-power reduction. The improved wing geometry validated under outdoor conditions achieved this 33% power reduction at $10\,\mathrm{m\,s^{-1}}$ compared with hover. These findings close the loop between aerodynamic theory, thrust-based proxy estimates, and real-flight measurements, providing the first quantitative evidence of efficiency improvement in an INDI-controlled aerodynamic multirotor.

The results confirm that disturbance-rejection control can absorb quasi-steady aerodynamic effects without identification or model compensation. This finding simplifies control design for hybrid multirotors: aerodynamic augmentation can be pursued purely at the hardware level, leaving the control law unchanged.

The main limitations are the absence of direct power measurements and quantitative comparison to a geometric-only baseline. Future work should instrument the system with current sensing to quantify energy savings, and perform controlled A/B tests to isolate the INDI improvement. Extending the framework to dynamic flight envelopes and real-time aerodynamic estimation would further test its generality. Additional future work includes:

- Extend tests to longer endurance missions to quantify range increase and include varying payload configurations.

Overall, the study establishes a reproducible experimental benchmark for aerodynamic surface-enhanced multirotors and provides evidence that model-free incremental control remains robust even when aerodynamic lift dominates vehicle dynamics.

A. Supplementary Material

A.1. Detailed Bill of Materials

This section provides a complete component list for the experimental X-wing quadrotor platform, including part numbers, specifications, and suppliers for reproducibility.

A.1.1. Platform Variant Comparison

Table A.2 provides a detailed comparison between the baseline platform developed in this work and the improved variant developed by Pablo Kirby.

A.2. Thrust Characterization Details

A.2.1. Test Procedure

Static thrust measurements were performed using the following procedure:

1. A single T-MOTOR VELOX V2808 motor with HQProp 7×3.5×3 propeller was mounted on a Rokubi Mini six-axis force–torque sensor using a custom 3D-printed fixture.
2. The motor was powered through the same flight controller (Kakute H7) and ESC configuration used in flight to ensure identical DShot timing and behavior.
3. Battery configuration: Two 6S 1400 mAh Li-Po batteries connected in parallel (6S 2P), matching the flight configuration.
4. Motor commands were swept from 0% to 100% throttle in 1% increments.
5. At each command level, the motor was allowed to stabilize for 1 s, then force data were recorded for 2 s and averaged.
6. The test was repeated with all four motors installed on the vehicle (three fixed, one on force sensor) to capture realistic voltage sag effects.
7. A total of 975 samples were recorded across the full throttle range.

A.2.2. Power Configuration Considerations

During the bench tests, we observed that the command-to-thrust relationship depends on how many motors are simultaneously powered from the battery due to load-dependent voltage sag and internal resistance effects.

In particular, running only a single motor from a single 6S pack produced a slightly different mapping than powering all four motors from the flight-representative configuration with two 6S packs in parallel (6S 2P). The voltage sag under multi-motor load shifts the thrust curve downward by approximately 5–8% at high throttle.

Unless stated otherwise, the mapping used for control in flight was calibrated under the latter 6S 2P, four-motor-powered condition, and all subsequent thrust conversions refer to this configuration.

A.3. Software Architecture Details

A.3.1. System Architecture

The onboard software stack consists of ROS 1 Noetic running on the Jetson Orin Nano companion computer. The control architecture is divided into three main components:

1. **Ground Station** (Remote PC):
    - ROS master node
    - Trajectory generation and mission planning
    - Real-time visualization (RViz)
    - Manual override interface
    - Data logging (rosbag)
2. **Companion Computer** (Jetson Orin Nano):
    - Geometric outer loop controller (300 Hz)
    - INDI inner loop controller (300 Hz)
    - State estimation (Vicon feedback processing)
    - MAVLink communication interface
    - Local telemetry logging
3. **Flight Controller** (Kakute H7):
    - Modified Betaflight firmware
    - IMU data acquisition (8 kHz)

- Motor RPM telemetry (DShot bidirectional)
- Battery voltage monitoring
- Motor command execution (DShot600, 8 kHz update rate)

A.3.2. Communication Protocols

Jetson ↔ Ground Station:

- Protocol: ROS 1 TCP/UDP
- Medium: Wi-Fi 802.11ac (5 GHz)
- Bandwidth: ≈50 Mbps sustained
- Latency: 5–15 ms (local network)

Jetson ↔ Kakute H7:

- Protocol: MAVLink 2
- Medium: UART (serial)
- Baud rate: 921 600 bit/s
- Update rate: 300 Hz (synchronized with control loop)
- Latency: <3 ms

A.3.3. State Estimation Pipeline

The state estimation pipeline used the Vicon motion capture system to provide real-time position and attitude feedback to the controller. In some experiments, an onboard Extended Kalman Filter (EKF) fused IMU data for state estimation backup.

The Vicon system and onboard IMU operated in a consistent right-handed coordinate convention:

- **X-axis**: Forward (red in figures)
- **Y-axis**: Left
- **Z-axis**: Up
- **Euler angles**: Roll (ϕ) about X, Pitch (θ) about Y, Yaw (ψ) about Z

A.3.4. Control Implementation

All flight maneuvers, including point-to-point transitions and trajectory tracking, were executed fully autonomously from predefined setpoints. The control logic ran entirely on the Jetson companion computer, with the flight controller acting only as a sensor interface and motor command relay.

Manual override from the ground station was available for safety. Trajectories could be manually stopped and the vehicle landed or disarmed from the base station using a dedicated emergency stop button.

A.4. Test Facility Details

A.4.1. Munich Indoor Arena

- **Dimensions**: Approximately 7×8 m flying volume
- **Vicon system**: 8 cameras
- **Sampling rate**: 200 Hz
- **Position accuracy**: ± 2 mm
- **Safety features**: Safety nets, soft landing surface
- **Use cases**: Agility experiments (3 g circle maneuvers, aggressive transitions)

A.4.2. AIDA Hall (Reutlingen University)

- **Dimensions**: Approximately 60×45 m flying volume
- **Vicon system**: 24 cameras
- **Sampling rate**: 200 Hz
- **Position accuracy**: ± 2 mm
- **Environment**: Indoor, still-air conditions, no wind sources
- **Use cases**: Steady forward-flight efficiency trials, high-speed tests (up to 20 m/s)

A.4.3. Test Protocol Details

Pre-flight:

1. Battery charge verification (>95% capacity)
2. Vicon marker visibility check
3. Ground station connection test
4. Emergency stop procedure rehearsal
5. Trajectory parameter verification

Flight sequence:

1. Autonomous take-off to 1.5 m altitude
2. Hover stabilization (10 s)
3. Trajectory execution (step response, circle, or straight line)
4. Return to hover
5. Autonomous landing

Post-flight:

1. Data integrity check (rosbag verification)
2. Timestamp synchronization validation
3. Outlier detection (position jumps, IMU spikes)
4. Visual trajectory inspection
5. Battery voltage log review

Acceptance criteria:

- No Vicon position jumps >50 mm
- No IMU saturation events
- Timestamp monotonicity preserved

- No external disturbances observed (visual confirmation)
- Consistent trajectory shape across repeated runs

Each experiment was repeated 3–5 times until these criteria were met for at least 3 consecutive runs. Only the best-quality datasets were used for analysis.

A.5. Data Processing Pipeline

All experiment data were logged in rosbag format and post-processed using custom Python scripts. The processing pipeline consisted of:

1. **Data extraction**: rosbag $\rightarrow$ CSV conversion using `rosbag_pandas`
2. **Timestamp alignment**: NTP-synchronized timestamps merged across Vicon and onboard telemetry
3. **Filtering**: Low-pass Butterworth filter (40 Hz cutoff) applied to Vicon velocity estimates
4. **Thrust estimation**: Motor RPM $\rightarrow$ thrust conversion using calibrated mapping
5. **Coordinate transformations**: Body frame $\leftrightarrow$ world frame conversions using measured attitude
6. **Error calculation**: Trajectory tracking errors computed as $\|\mathbf{p}_d - \mathbf{p}\|$
7. **Statistical analysis**: Mean, RMS, max error, standard deviation computed for each run
8. **Visualization**: Matplotlib/pgfplots for publication-quality figures

All processing scripts are documented in the `scripts/` directory of the thesis repository.

A.6. Manufacturing Details

A.6.1. 3D Printing Parameters

Wing skins (PLA Aero):

- **Printer**: Bambu Lab X1-Carbon

- **Material**: Bambu Lab PLA Aero (lightweight variant)
- **Print mode**: Vase mode (single-wall spiral)
- **Layer height**: 0.2 mm
- **Wall thickness**: 0.4 mm (single perimeter)
- **Infill**: 0% (hollow structure)
- **Flow ratio**: 0.6
- **Print speed**: 100 mm/s
- **Nozzle temperature**: 245°C
- **Bed temperature**: 60°C
- **Support**: None required
- **Print time**: ≈2.5 h per wing

Frame components (standard PLA):

- **Printer**: Bambu Lab X1-Carbon
- **Material**: Bambu Lab PLA Basic (black)
- **Layer height**: 0.2 mm
- **Wall lines**: 4 (1.6 mm total)
- **Infill**: 20% grid
- **Print speed**: 150 mm/s
- **Nozzle temperature**: 220°C
- **Bed temperature**: 60°C
- **Support**: Tree supports (auto-generated)

Table A.1.: Complete Bill of Materials (BOM) for the baseline experimental platform.

Category	Component	Model / Key Specs
Airframe		
Frame	Center frame	X-wing center frame, 3D-printed PLA
Frame	Wing spars	Carbon fiber tubes, 10 mm OD, 8 mm ID
Wings	Wing skins	Four fixed wings (length 450 mm, chord 250 mm; NACA 0015 airfoil), orthogonal layout (90° between adjacent wings); 3D-printed Bambu Lab PLA Aero; total projected horizontal area $S_t \approx 0.318\,\mathrm{m}^2$
Propulsion		
Motors	T-MOTOR VELOX V2808 (×4)	KV1300, max thrust ≈22 N each @ 6S
Propellers	HQProp 7×3.5×3 (×4)	3-blade racing propeller (7", light grey)
Avionics		
Flight Controller	Kakute H7 v1.5	STM32H743, ICM-42688-P IMU @ 8 kHz
ESC	Holybro Tekko32 F4 4in1	4-in-1 ESC, 65A, DShot600/1200 capable
Power		
Battery (×2, parallel)	Tattu R-Line V5.0 6S 1400 mAh	150C discharge, XT60 connector; effective 6S 2800 mAh (parallel)
Companion Power	Tattu R-Line V3.0 4S 1800 mAh	120C discharge, XT60 connector
Companion Computer		
Computer	NVIDIA Jetson Orin Nano 8 GB	Ubuntu 20.04, ROS 1 Noetic
Motion Capture		
Markers	Vicon markers (×4)	14 mm diameter, asymmetric constellation

Table A.2.: Detailed comparison of baseline and improved X-wing platform variants.

Parameter	Baseline Platform (This Work)	Improved Platform (Kirby, 2025)
Aerodynamics		
Airfoil	NACA 0015 (symmetric)	AG25 (cambered)
Dihedral/Anhedral	$\pm 45^\circ$ (alternating)	$\pm 20^\circ$ (alternating)
Wing Span	450 mm	500 mm
Wing Chord	250 mm	230 mm
Wing Thickness	37.5 mm (15% chord)	57.5 mm (25% chord)
Total Wing Area	0.318 m^2	0.345 m^2
Aspect Ratio (per wing)	1.8	2.17
Structural		
Wing Material	Bambu Lab PLA Aero	Bambu Lab PLA Aero
Frame Material	Standard PLA	Standard PLA
Motor Tilt	$\pm 5^\circ$	$\pm 5^\circ$
Total Mass	2.5 kg	2.8 kg
Propulsion	Identical	Identical
Motors	T-MOTOR VELOX V2808 (×4)	T-MOTOR VELOX V2808 (×4)
Propellers	HQProp 7×3.5×3 (×4)	HQProp 7×3.5×3 (×4)
Battery	6S 2P 2800 mAh	6S 2P 2800 mAh
Avionics	Identical	Identical
Flight Controller	Kakute H7 v1.5	Kakute H7 v1.5
Companion	Jetson Orin Nano 8 GB	Jetson Orin Nano 8 GB
Control	Identical	Identical
Control Architecture	INDI with aerodynamic compensation	INDI with aerodynamic compensation
Control Frequency	300 Hz inner & outer	300 Hz inner & outer

Abbreviations

UAV Unmanned Aerial Vehicle
MAV Micro Air Vehicle
VTOL Vertical Take-Off and Landing
INDI Incremental Nonlinear Dynamic Inversion
EoM Equations of Motion
DoF Degrees of Freedom
CoG Center of Gravity
CoP Center of Pressure
IMU Inertial Measurement Unit
Vicon Vicon Motion Capture System
FC Flight Controller
ESC Electronic Speed Controller
RPM Revolutions per Minute

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